

Probing Dark Matter Substructures with Free-Form Modelling: A Case Study of the ‘Jackpot’ Strong Lens

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ABSTRACT

Characterising the population and internal structure of sub-galactic halos is critical for constraining the nature of dark matter. These halos can be detected near galaxies that act as strong gravitational lenses with extended arcs, as they perturb the shapes of the arcs. However, this method is subject to false-positive detections and systematic uncertainties, particularly degeneracies between an individual halo and larger-scale asymmetries in the distribution of lens mass. We present a new free-form lens modelling code, developed within the framework of the open-source software PyAutoLens, to address these challenges. Our method models mass perturbations that cannot be captured by parametric models as pixelized potential corrections and suppresses unphysical solutions via a Matérn regularisation scheme that is inspired by Gaussian process regression. This approach enables the recovery of diverse mass perturbations, including subhalos, line-of-sight halos, external shear, and multipole components that represent the complex angular mass distribution of the lens galaxy, such as boxiness/diskiness. Additionally, our fully Bayesian framework objectively infers hyperparameters associated with the regularisation of pixelized sources and potential corrections, eliminating the need for manual fine-tuning. By applying our code to the well-known ‘Jackpot’ lens system, SLACS0946+1006, we robustly detect a highly concentrated subhalo that challenges the standard cold dark matter model. This study represents the first attempt to independently reveal the mass distribution of a subhalo using a fully free-form approach.

Key words: gravitational lensing; strong — galaxies: individual: SLACS0946+1006 — dark matter

1 INTRODUCTION

The standard cosmological model has achieved remarkable success (Planck Collaboration et al. 2020), as it explains numerous observations, such as the rotation curves of disk galaxies (Rubin 1991), the clustering properties of galaxies (Springel et al. 2005), the properties of the hot gas in galaxy clusters (Ettori et al. 2013), and the cosmic microwave background (Planck Collaboration et al. 2014). The standard cosmological model assumes that approximately 85 percent of the matter in the universe is the so-called “dark matter” (Planck Collaboration et al. 2020). Dark matter neither emits light nor interacts electromagnetically with baryonic matter. Instead, its presence is inferred solely from gravitational effects (Arbey & Mahmoudi 2021).

The fundamental nature of dark matter remains elusive; it may be primordial black holes (Carr et al. 2016) or some undiscovered particles beyond the standard model of particle physics (Feng 2010). Since different types of dark matter produce distinct cosmological structures, such as dark matter halos and subhalos, astronomical observations can serve as an effective means to constrain the properties of dark matter (Frenk & White 2012). The popular probes currently used to infer the properties of dark matter include stellar streams in the local group (Ibata et al. 2001; Bonaca et al. 2019), the Lyman-alpha forest (Seljak et al. 2006), the luminosity function of Milky Way satellite galaxies (Nadler et al. 2021; Newton et al. 2021), and strong gravitational lensing (Vegetti et al. 2023). Among these, the strong gravitational lensing technique offers a distinct advantage, as it can infer the properties of dark matter solely based on the gravitational distortion effects of dark matter structures, without relying on complex baryonic physics, and it can be applied to higher redshifts.

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Strong gravitational lensing occurs when light from a background source is significantly distorted by a massive foreground deflector, such as a galaxy or a galaxy cluster, forming multiple images or extended arcs (Schneider et al. 1992; Meylan et al. 2006; Meneghetti 2022). Smaller dark matter halos within the main deflector (subhalos) or along the line of sight (“LOS-halos”, Li et al. 2017; Despali et al. 2018; He et al. 2022b) can further perturb the lensing signal, creating observable anomalies in lensed images, time delays, and flux ratios between multiple images. These anomalies serve as valuable probes for constraining the properties of dark matter (Vegetti & Koopmans 2009b; Li et al. 2016; He et al. 2022a; Keeton & Moustakas 2009; Liao et al. 2018; Mao & Schneider 1998; Birrer et al. 2017; Gilman et al. 2018; Hsueh et al. 2020). Current time-delay measurements, typically with uncertainties on the order of a few days (Millon et al. 2020), lack the precision required to detect subhalo-induced signals, which are often on the order of fractions of a day. Flux ratio anomalies, theoretically most sensitive to subhalo perturbations due to their direct dependence on the second derivative of the lensing potential, generally cannot precisely constrain the properties (e.g., mass and position) of individual subhalos (e.g. Nierenberg et al. 2014). This limitation arises from the sparsity of observational data (usually restricted to positions and fluxes of lensed images), which does not provide sufficient constraints to resolve degeneracies either among subhalo internal properties (e.g., mass-position degeneracy) or between the subhalo and the main lens galaxy (Fadely & Keeton 2012; Hsueh et al. 2020; Cohen et al. 2024). Consequently, subhalo properties inferred from flux ratio anomalies are typically derived statistically—finding multiple feasible solutions across a range of subhalo masses and positions—instead of precise individual measurements (Dalal & Kochanek 2002; Hsueh et al. 2020; Gilman et al. 2020). In contrast, the brightness distribution along distorted lensed arcs, typically comprising several thousand pixels, can effectively break the degeneracies encountered by flux ratio anomalies. Thus, analysing extended arcs offers a more reliable method for precisely measuring the properties of individual subhalos through strong gravitational lensing (e.g., Vegetti et al. 2010, 2012; Hezaveh et al. 2016; Ritondale et al. 2019; Nightingale et al. 2024).

Subhalo detections using distorted lensed arcs have been widely employed to constrain dark matter properties (e.g., Vegetti et al. 2014, 2018; Enzi et al. 2021). Since the distortions induced by small-mass subhalos are subtle, sophisticated lens modelling methods are essential for extracting these weak signals. Currently, two primary approaches are employed: parametric and free-form methods. The parametric method uses explicit parametric models to describe the mass distribution of both the main lens galaxy and the presumed subhalo. In this approach, a subhalo is considered detected if a model including a subhalo provides a significantly better fit to the observational data, as measured by an increase in Bayesian evidence, compared to a model without a subhalo (Vegetti & Koopmans 2009a; Ritondale et al. 2019; Nightingale et al. 2024). In contrast, the free-form method approximates the mass distribution of the main lens galaxy using a smooth parametric model while treating perturbative signals not captured by the parametric model as pixelized linear corrections to the lensing potential¹ (Koopmans 2005; Suyu & Blandford 2006; Suyu et al. 2009; Vegetti & Koopmans 2009a; Ritondale et al. 2019;

Vernardos & Koopmans 2022; Galan et al. 2022; Rüstig et al. 2024). Subhalos appear as localised positive mass clumps in the resulting mass perturbation map. While the parametric method is effective at quantifying the properties of subhalo perturbers, it is prone to false positives, where a subhalo is incorrectly inferred to exist (Ritondale et al. 2019). This issue stems from systematic errors in the parametric model, particularly degeneracies between the mass distribution of the main lens galaxy and the subhalo. If the main lens galaxy’s mass model is oversimplified, a false-positive subhalo signal may arise to compensate for missing complexity in the lens galaxy (He et al. 2023; Nightingale et al. 2024; O’Riordan & Vegetti 2024). Although adding complexity to the lens mass model can test the robustness of subhalo detections (Lange et al. 2024), it is often unclear whether the added complexity is sufficient to eliminate all false positives. Consequently, subhalo detection claims based solely on parametric methods may be met with scepticism. To address this concern, Vegetti et al. (2014) advocated for free-form approaches to validate subhalo detections obtained using parametric methods.

Regularisation, or smoothing, is essential for obtaining physically meaningful solutions in free-form approaches (Wallington et al. 1996; Warren & Dye 2003). The classical potential correction method begins with an initial estimate of the main lens mass distribution and the source brightness distribution. A strong regularisation strength is empirically applied to the pixelized lensing potential perturbations, producing a minor correction to the lensing potential (Koopmans 2005; Vegetti & Koopmans 2009a). This correction is subsequently used to refine both the lens mass and source light models, initiating a new iteration of potential correction. The process is repeated iteratively until convergence is achieved, defined as the point where additional corrections no longer result in significant improvements to the model fit (Koopmans 2005). A major limitation of this iterative algorithm is its vulnerability to instability: artefacts caused by an inappropriate choice of regularisation strength can accumulate over iterations, ultimately compromising the modelling process. Vernardos & Koopmans (2022) introduced a framework to objectively determine the regularisation strength for pixelized lensing potential perturbations by maximising Bayesian evidence and directly solving for the potential correction, thereby avoiding the need for unstable iterative algorithms. However, the Gaussian and Exponential regularisation techniques used in their study are more effective for extended perturbations but perform poorly when applied to localised features, such as those caused by subhalos².

The Matérn kernel is a widely used covariance function that imposes smoothness constraints in free-form reconstruction via Gaussian process regression. This Matérn covariance kernel, or regularisation, has recently been applied to pixelised source reconstruction in strong gravitational lensing analyses (Galan et al. 2024; Enzi et al. 2024; Rüstig et al. 2024). Whilst Vernardos & Koopmans (2022) primarily investigated the advantages of Gaussian and exponential forms of regularisation—special cases of the more general Matérn kernel—for reconstructing pixelised sources and mass models, they also proposed that the Matérn kernel merits further exploration. In this work, we present a new free-form lens modelling code built upon the open-source software PyAutoLens (Nightingale & Dye 2015; Nightingale et al. 2021, 2018). By employing a generalised Matérn regularisation for the pixelised mass model (in the form of potential corrections) (Rasmussen & Williams 2006) and determining the regularisation parameters objectively through the Bayesian evidence, our code automatically recovers diverse forms of mass per-

¹ This free-form approach to the pixelized perturbative lensing potential is widely known as the potential correction method. Other types of free-form approaches have also been employed in strong lensing analyses, which represent the mass (instead of the lensing potential) on a grid of pixels (e.g. Saha & Williams 1997) or as a linear combination of basis functions (e.g. Liesenborgs et al. 2006; Diego et al. 2007).

² Further demonstration of this limitation is provided in Section 4.1.

turbation signals without requiring manual fine-tuning. To validate our method, we first test it on mock lensing systems, demonstrating its robustness in recovering various forms of lensing potential perturbations commonly encountered in strong lensing studies. We then apply our method to the strong lens system SLACS0946+1006 (Gavazzi et al. 2008; Smith & Collett 2021), also known as the ‘Jackpot’ lens, and characterise the mass perturbations in this system using a fully free-form approach.

This paper is organised as follows. In Section 2, we describe the methodology for the potential correction. Section 3 describes the datasets employed in this study, including mock datasets and the real “Jackpot” lens system (SLACS0946+1006). In Section 4, we first evaluate the performance of our method in quantitatively recovering mass perturbation signals using mock datasets and then present a free-form analysis of the real lens system SLACS0946+1006. The strengths, limitations, and future prospects of our method are discussed in Section 5, and a concise summary is provided in Section 6. Unless otherwise stated, we adopt the posterior median as the default point estimate for the model parameters. Throughout this work, we assume a flat Λ CDM cosmology with $\Omega_m = 0.3$, $\Omega_\Lambda = 0.7$, and $H_0 = 70 \text{ km s}^{-1} \text{ Mpc}^{-1}$. All codes and relevant data used in this work are publicly available at the following link: https://github.com/caoxiaoyue/lensing_potential_correction.

2 METHODOLOGY

This section provides a concise review of the framework for the potential correction method. Section 2.1 presents the linear approximation that underpins the potential correction method. Section 2.2 outlines the framework for only solving lensing potential perturbations given an initial estimate of the lens mass and source. Section 2.3 describes how to simultaneously solve for source and lensing potential perturbations, accounting for their covariance. The configuration of the free-form model, including the pixelization and regularisation strategies, is presented in Section 2.4.

2.1 Linear Approximation Relation

In galaxy-galaxy strong lens modelling, a smooth parametric mass model for the lens galaxy, such as the power-law plus shear model, typically provides a satisfactory fit to the global morphological features of the lensing image (e.g. Shajib et al. 2021; Etherington et al. 2022; Tan et al. 2024). Residuals between the model and data images are usually small, suggesting that any omitted mass complexities unaccounted for in this smooth lens model (commonly referred to as the macro model) are minor and can be treated as small perturbations to the lensing potential. The potential correction method aims to identify a set of such perturbations defined on a pixelized grid, reducing image residuals to a level consistent with observational noise (Koopmans 2005). Such perturbations can be identified using a linear approximation that relates the lensing potential perturbation, the source brightness gradient, and the image residual, as derived next.

Let $R^L(\theta)$ denote the residuals in the image produced by the macro model at the image-plane angular position θ , defined as

$$R^L(\theta) \equiv I^T(\theta) - I^M(\theta) + N(\theta). \quad (1)$$

Here, I^T denotes the true brightness of the lensed images, and I^M denotes the brightness predicted by the macro model. $N(\theta)$ represents the image noise realisation. The residual $R^L(\theta)$ is related to

the intrinsic true source brightness distribution, S , via

$$R^L(\theta) - N(\theta) = \text{PSF} \otimes [S(\beta^T) - S(\beta^M)], \quad (2a)$$

$$= \text{PSF} \otimes \left[\frac{\partial S}{\partial \beta} (\beta^T - \beta^M) \right], \quad (2b)$$

where PSF denotes the point spread function kernel, and $\otimes$ represents the convolution operator. The quantities β^T and β^M indicate the ray-traced source-plane positions corresponding to θ under the lensing potentials given by the ground truth (ψ^T) and the macro model (ψ^M), respectively. Explicitly,

$$\beta^T = \theta - \frac{\partial \psi^T(\theta)}{\partial \theta}, \quad (3a)$$

$$\beta^M = \theta - \frac{\partial \psi^M(\theta)}{\partial \theta}, \quad (3b)$$

$$\psi^T(\theta) \equiv \psi^M(\theta) + \delta\psi(\theta), \quad (3c)$$

where $\delta\psi$ denotes a small difference between the lensing potential of the macro model and the ground truth. It follows that

$$\beta^T - \beta^M = \frac{\partial (\psi^M - \psi^T)}{\partial \theta} = -\frac{\partial \delta\psi}{\partial \theta}. \quad (4)$$

Equation 2a implicitly assumes that the image residual R^L arises purely from the small displacements in the deflection angle caused by $\delta\psi$. This is a reasonable approximation, given that $\delta\psi$ is a small quantity. Substituting Equation 4 into Equation 2b yields

$$R^L(\theta) = \text{PSF} \otimes \left[\frac{\partial S}{\partial \beta} \cdot \left(-\frac{\partial \delta\psi}{\partial \theta} \right) \right] + N(\theta). \quad (5)$$

This result demonstrates that the image residuals are proportional to the product of the source brightness gradient and the gradient of the lensing potential perturbation. Note that in practical analyses, the intrinsic true source brightness distribution, S , is generally unknown and is typically approximated by the macro model’s prediction.

2.2 Inverting Only the Potential Perturbation

The primary goal of the potential correction algorithm is to solve the partial differential Equation 5, using the image residual and source gradient from the macro model. A straightforward method for solving this equation is numerical integration along the characteristic curve, as outlined in Suyu & Blandford (2006). However, this approach is effective only when the perturbative lensing potential is small (less than 1%). A more robust alternative for solving equation 5 is to reformulate it as the following matrix expression:

$$\begin{aligned} \delta d &= -\mathbf{B} \mathbf{D}_s \mathbf{C}_\psi \mathbf{D}_\psi \delta\psi + \mathbf{n} \\ &= \mathbf{L}_\psi \delta\psi + \mathbf{n}. \end{aligned} \quad (6)$$

We explain the meaning of each term in equation 6 as follows:

(i) δd is a column vector where each element δd_j represents the value of image residuals provided by the macro model for each image pixel, where $j = 1, \dots, N_d$, and N_d denotes the number of image pixels. Similarly, the vector $\mathbf{n}$ represents the noise realisation at each image pixel, with values drawn from the multivariate normal distribution $\mathcal{N}(0, \mathbf{C}_D)$, where $\mathbf{C}_D$ is the noise covariance matrix.

(ii) $\mathbf{B}$ is a square matrix of dimensions $[N_d, N_d]$ constructed to account for the blurring effect of the PSF.

(iii) The vector $\delta\psi$ comprises elements $\delta\psi_k$, each representing the value of the lensing potential perturbation at the k -th potential correction grid, where $k = 1, \dots, N_p$, and N_p denotes the number of pixels in the grid. The potential correction grid is typically coarser

than the native image grid, such that $N_p < N_d$, to enable a more efficient inversion process.

(iv) $\mathbf{D}_\psi$ is a gradient operation matrix (dimensions: $2N_p \times N_p$) which, when acting on $\delta\psi$, generates a column vector (dimensions: $2N_p \times 1$) representing the gradient of $\delta\psi$. The generic structure of the matrix $\mathbf{D}_\psi \delta\psi$ is shown below:

$$\mathbf{D}_\psi \delta\psi = \begin{pmatrix} \dots \\ \frac{\partial \delta\psi(\theta_j^p)}{\partial \theta_x} \\ \frac{\partial \delta\psi(\theta_j^p)}{\partial \theta_y} \\ \frac{\partial \delta\psi(\theta_{j+1}^p)}{\partial \theta_x} \\ \frac{\partial \delta\psi(\theta_{j+1}^p)}{\partial \theta_y} \\ \dots \end{pmatrix}. \quad (7)$$

$\frac{\partial \delta\psi(\theta_j^p)}{\partial \theta_x}$ and $\frac{\partial \delta\psi(\theta_j^p)}{\partial \theta_y}$ represent the x and y components of the gradient of the potential perturbation at the j -th potential correction grid, denoted by θ_j^p .

(v) $\mathbf{C}_f$ is an interpolation matrix with dimensions of $N_d \times N_p$ that maps a column vector defined on the coarser ‘‘potential correction grid’’ to a new vector defined at the native image resolution through linear interpolation.

(vi) $\mathbf{D}_s$ is a matrix with dimensions $N_d \times 2N_d$ that stores the gradient of the source surface brightness. The generic structure of the matrix $\mathbf{D}_s$ is shown below:

$$\mathbf{D}_s = \begin{pmatrix} \dots & \frac{\partial S(\beta_j^d)}{\partial \beta_x} & \frac{\partial S(\beta_j^d)}{\partial \beta_y} & \dots \\ \dots & \frac{\partial S(\beta_{j+1}^d)}{\partial \beta_x} & \frac{\partial S(\beta_{j+1}^d)}{\partial \beta_y} & \dots \end{pmatrix}. \quad (8)$$

β_j^d represents the ray-traced source plane position of the j -th native image grid (with position denoted by θ_j^d), according to the macro model. Let S be the source brightness distribution predicted by the macro model. The x and y components of the source’s brightness gradient are given by $\frac{\partial S(\beta_j^d)}{\partial \beta_x}$ and $\frac{\partial S(\beta_j^d)}{\partial \beta_y}$.

For convenience, we abbreviate $-\mathbf{B}\mathbf{D}_s\mathbf{C}_f\mathbf{D}_\psi$ as $\mathbf{L}_{\delta\psi}$. The latter has dimensions of $N_d \times N_p$.

It is evident that $\delta\mathbf{d}$ and $\delta\psi$ are connected via a linear mapping. Following the classical ‘semi-linear inversion’ framework of Warren & Dye (2003), the maximum a posteriori estimate for $\delta\psi$, for a given $\mathbf{L}_{\delta\psi}$ and regularisation $\mathbf{R}_{\delta\psi}$, can be obtained via

$$\delta\psi_{\text{MAP}} = \left(\mathbf{L}_{\delta\psi}^T \mathbf{C}_D^{-1} \mathbf{L}_{\delta\psi} + \mathbf{R}_{\delta\psi} \right)^{-1} \mathbf{L}_{\delta\psi}^T \mathbf{C}_D^{-1} \delta\mathbf{d}. \quad (9)$$

Here, the regularisation matrix $\mathbf{R}_{\delta\psi}$ imposes a covariance prior on the pixelised potential correction $\delta\psi$, thereby reducing the effective degrees of freedom and avoiding ill-posed solutions. The optimal regularisation $\mathbf{R}_{\delta\psi}$ can be determined by maximising the Bayesian evidence $\mathcal{E}$, whose formula was derived by Suyu et al. (2006) and reorganised by Dye et al. (2008), as follows:

$$\begin{aligned} \log \mathcal{E} = & -\frac{N_d}{2} \log(2\pi) - \frac{1}{2} \log(\det \mathbf{C}_D) \\ & + \frac{1}{2} \log(\det \mathbf{R}_{\delta\psi}) - \frac{1}{2} \delta\psi_{\text{MAP}}^T \mathbf{R}_{\delta\psi} \delta\psi_{\text{MAP}} \\ & - \frac{1}{2} (\mathbf{L}_{\delta\psi} \delta\psi_{\text{MAP}} - \delta\mathbf{d})^T \mathbf{C}_D^{-1} (\mathbf{L}_{\delta\psi} \delta\psi_{\text{MAP}} - \delta\mathbf{d}) \\ & - \frac{1}{2} \log[\det(\mathbf{L}_{\delta\psi}^T \mathbf{C}_D^{-1} \mathbf{L}_{\delta\psi} + \mathbf{R}_{\delta\psi})]. \end{aligned} \quad (10)$$

A more thorough description of the full Bayesian framework can be found in Appendix A.

2.3 Simultaneous Inversion of Source and Potential Perturbations

In Section 2.2, we fix the source light to the result given by the macro model, fitting only for $\delta\psi$ using equation 6. However, this process does not account for the covariance between the source light and $\delta\psi$. It has been shown that the source light model can absorb the image residual left by the imperfect lens mass model (e.g. Vernardos & Koopmans 2022; Nightingale et al. 2024). Therefore, a better approach is to fit the source light and $\delta\psi$ simultaneously, allowing them to compete to reconstruct the observed lensing image. As demonstrated by Koopmans (2005); Vegetti & Koopmans (2009a), this can be achieved with simple manipulations on equation 6. Suppose we use a pixelized source model for the macro model, then the image residual left by the macro model, which appears on the left-hand side of equation 6, is equal to

$$\delta\mathbf{d} = \mathbf{d} - \mathbf{B}\mathbf{L}_s(\psi_p)\mathbf{s}. \quad (11)$$

Each element d_j in the vector $\mathbf{d}$ represents the brightness value of the lensing image at each pixel. Each element s_i in the vector $\mathbf{s}$ represents the brightness value at each source pixel. $\mathbf{L}_s(\psi_p)$ is the lens mapping matrix with dimensions $[N_d, N_s]$, which, when acting on $\mathbf{s}$, generates the lensed source image. The value of $\mathbf{L}_s(\psi_p)$ depends on the macro model’s lens mass model (denoted by its lensing potential ψ_p) and the interpolation scheme of the pixelized source model. For a more thorough illustration of how to construct this matrix $\mathbf{L}_s(\psi_p)$, refer to Appendix B of Treu & Koopmans (2004). On the right-hand side of equation 6, the construction of the matrix $\mathbf{L}_{\delta\psi}$ requires knowledge of the source brightness distribution S , which can be obtained by interpolating the pixelized source reconstruction (denoted by s_p), given by the macro model. Reorganizing equation 6, we have

$$\begin{aligned} \mathbf{d} &= \mathbf{B}\mathbf{L}_s(\psi_p)\mathbf{s} + \mathbf{B}\mathbf{L}_{\delta\psi}(s_p)\delta\psi + \mathbf{n} \\ &= \mathbf{B}\mathbf{L}(\psi_p, s_p)\mathbf{r} + \mathbf{n}. \end{aligned} \quad (12)$$

Where we introduce the block matrices

$$\mathbf{L}(\psi_p, s_p) \equiv \left(\mathbf{L}_s(\psi_p) \mid \mathbf{L}_{\delta\psi}(s_p) \right), \quad (13)$$

and

$$\mathbf{r} \equiv \begin{pmatrix} \mathbf{s} \\ \delta\psi \end{pmatrix}. \quad (14)$$

It is immediately clear that the matrix equation 12 is linear; therefore, all the mathematical frameworks shown in Section 2.2 to solve Equation 6 can be applied to Equation 12. All we need to do is to replace $\delta\mathbf{d}$ with $\mathbf{d}$, $\mathbf{L}_{\delta\psi}$ with $\mathbf{L}$, and $\delta\psi$ with $\mathbf{r}$ (see also Vernardos & Koopmans 2022). One issue worth noting is that the regularisation prior $\{\mathbf{g}_r, \xi_r\}$ for the vector $\mathbf{r}$ can be split into two independent parts: the source light regularisation $\{\mathbf{g}_s, \xi_s\}$ and the lensing potential perturbation regularisation $\{\mathbf{g}_{\delta\psi}, \xi_{\delta\psi}\}$. The regularisation functions for both $\mathbf{s}$ and $\delta\psi$ also take a quadratic form, expressed as

$$\begin{aligned} E_s(\mathbf{s}) &= \frac{1}{2} \mathbf{s}^T \mathbf{R}_s \mathbf{s}, \\ E_{\delta\psi}(\delta\psi) &= \frac{1}{2} \delta\psi^T \mathbf{R}_{\delta\psi} \delta\psi. \end{aligned} \quad (15)$$

Here, $\mathbf{R}_s$ and $\mathbf{R}_{\delta\psi}$ represent the Hessians of the regularisation functions for the source (E_s) and the lensing potential perturbation ($E_{\delta\psi}$).

respectively. The regularisation function of $\mathbf{r}$ is

$$\begin{aligned} E_r(\mathbf{r}) &= E_s(s) + E_{\delta\psi}(\delta\psi) \\ &= \frac{1}{2} \mathbf{r}^T \mathbf{R}_r \mathbf{r} \end{aligned} \quad (16)$$

whose Hessian $\mathbf{R}_r$ can be further decomposed as

$$\mathbf{R}_r = \begin{pmatrix} \mathbf{R}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_{\delta\psi} \end{pmatrix}. \quad (17)$$

We note that solving the unknown perturbations in the lensing potential, whether based on equation 6 or 12, requires a non-zero source gradient (embedded in the $\mathbf{D}_s$ matrix). This implies that the potential correction method is, in principle, only applicable in the annular region where lensed emission is present. In practice, it is possible to use a larger mask that includes more regions without significant lensed emission and to let regularisation forcibly remove ill-posed solutions. However, as discussed in Section 5.1, this approach tends to be sub-optimal.

2.4 Configurations for the Free-Form Model

The free-form model discussed in this work represents model quantities using a pixelized field defined on a set of discretised grids, where each pixel's value is a free parameter. A free-form model includes three key components: pixelization, which determines the construction of the discretised grids; regularisation, which specifies how the field values across different pixels are correlated or smoothed; and the interpolation scheme, which describes how to map a pixelized field to a continuous one. In Sections 2.4.1 and 2.4.2, we present the configurations of free-form models for the source light and lensing potential perturbations, respectively. Their specific settings are summarised in Table 1.

2.4.1 Configurations for Source Light

In this work, we employ the **Overlay** model in PyAutoLens as the default pixelization scheme for the free-form source model. The **Overlay** scheme begins by overlaying a rectangular grid of size $n_x \times n_y$ pixels within the modelling area defined by the mask³. Subsequently, the grid of unmasked pixels is mapped to the source plane according to the current lens mass model, forming the source grid. The values of n_x and n_y are free parameters for the **Overlay** model, and their optimal values can be determined using Bayesian evidence. When the structure of a background source in a lens system is compact and clumpy, the **KMeans** model in PyAutoLens can be used to create grids for the free-form source model. This ensures that more source pixelisation generating points cluster in regions where the source is brighter, achieving the resolution needed to fit the lensing image. Specifically, the **KMeans** model uses the emission from the lensed source as a weight (by applying the k-means clustering algorithm) to randomly select N_s source pixels within the unmasked modelling area. This approach naturally generates a greater number of source pixels in regions with brighter lensed source emission. The grids on the image plane are then traced back to the source based on the current lens mass model to form the source grids. N_s can be either a free model parameter or fixed; the default setting in this work is to set N_s to half of the total number of image pixels within the unmasked annular region enclosing the lensed images.

³ The mask identifies the region of the lensing image **included** in the lens model.

The **ConstantSplit** model in PyAutoLens is employed as the default regularisation choice for the free-form source model. Essentially, this model regularises the brightness value of arbitrary source pixels, s_j , with all of their neighbouring pixels in a gradient fashion. Specifically, the regularisation function follows the form:

$$\begin{aligned} E_s(s) &= \sum_{j=1}^J \lambda_j^s \left\{ [s_j - \tilde{s}(x_j + l_j, y_j + l_j)]^2 \right. \\ &\quad + [s_j - \tilde{s}(x_j - l_j, y_j + l_j)]^2 \\ &\quad + [s_j - \tilde{s}(x_j + l_j, y_j - l_j)]^2 \\ &\quad \left. + [s_j - \tilde{s}(x_j - l_j, y_j - l_j)]^2 \right\}. \end{aligned} \quad (18)$$

Here, $\tilde{s}$ represents a continuous expression of the pixelized source model, derived using natural neighbour interpolation (Sibson 1981) under the Voronoi tessellation for the source grid. (x_j, y_j) denote the grid coordinates of the j -th source pixel, while l_j is the ‘‘cross size’’ of the j -th source pixel, calculated as the square root of the area of the j -th source pixel's Voronoi cell. The factor λ_j^s determines the overall regularisation (or smoothing) strength imposed on the j -th source pixel; for the **ConstantSplit** model, λ_j^s is a universal value across all source pixels. In cases where the source shows a high spatial variance in brightness distribution, regularisation schemes that allow for more dynamic range are needed. For faint regions on the source plane, we impose a higher regularisation strength to suppress noise, while for bright regions, we impose a lower regularisation to give the free-form model more freedom to reconstruct detailed morphological features of the source. The **AdaptiveBrightnessSplit** model in PyAutoLens provides such functionality. Specifically, this is done by directly ray-tracing the lensed source emission back to the source plane based on the current lens mass model and calculating the average brightness value A_j for the j -th source pixel. The A_j value is then mapped to the regularisation strength factor λ_j^s , such that a lower A_j value corresponds to a higher λ_j^s value.

For more detailed information on various source pixelization, regularisation, and interpolation schemes provided by PyAutoLens, we refer readers to Nightingale et al. (2018, 2021, 2024); He et al. (2024).

2.4.2 Configurations for Lensing Potential Perturbations

Solving the unknown lensing potential perturbation with Equation 6 or 12 requires knowledge of non-vanishing source gradient information, meaning the potential correction method only works in an annular region that tightly encloses the lensed source images. The pixelization grid for the free-form model of lensing potential perturbation is thus defined within this annular region, using a regular grid with a pixel size $N_{\delta\psi}$ times larger than that of the native lensing image⁴, where $N_{\delta\psi} = 1, 2, \dots$. The default $N_{\delta\psi}$ value used in this work is 2, which strikes a balance between computational efficiency and the accuracy required to recover the small mass perturbations induced by subhalos.

The Matérn regularisation (Stein 1999; Rasmussen & Williams 2006; Matérn 2013) is employed to control the smoothness of the pixelized lensing potential perturbation. For any two pixels of the lensing potential perturbation, denoted by their coordinates (x_m, y_m) and (x_n, y_n) , the correlation between them can be described by

$$C_{ij}(\rho, \nu) = \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\sqrt{2\nu} \frac{d_{ij}}{\rho} \right)^\nu K_\nu \left(\sqrt{2\nu} \frac{d_{ij}}{\rho} \right), \quad (19)$$

⁴ This functionality is supported by the **RegularDpsiMesh** module in our code, where the factor parameter corresponds to $N_{\delta\psi}$.

Free-Form Source Model			
Pixelization	Overlay	Shape	Fixed
	KMeans	Pixels	Fixed
		Weight Floor	$\mathcal{L}(10^{-5}, 1.0)$
		Weight Power	$\mathcal{U}(0.0, 5.0)$
regularisation	ConstantSplit	Coefficient	$\mathcal{L}(10^{-6}, 10^6)$
	AdaptiveBrightnessSplit	Inner Coefficient	$\mathcal{L}(10^{-6}, 10^6)$
		Outer Coefficient	$\mathcal{L}(10^{-6}, 10^6)$
		Signal Scale	$\mathcal{U}(0.0, 1.0)$
Interpolation	VoronoiNN	N/A	
Free-Form Model for Lensing Potential Perturbations			
Pixelization	RegularDpsiMesh	Factor	2
regularisation	MaternKernel	Coefficient	$\mathcal{L}(10^{-6}, 10^6)$
		Scale ["]	$\mathcal{L}(10^{-4}, 10^3)$
		Nu	$\mathcal{U}(0.5, 10.0)$
Interpolation	Bilinear Interpolation	N/A	

Table 1. Settings for the Free-Form Models of the Source and Lensing Potential Perturbations. From left to right, the table presents the components of the free-form model (pixelization, regularisation, or interpolation), the specific code modules employed, the parameter names of the code modules, and their prior settings. $\mathcal{U}$ and $\mathcal{L}$ denote uniform and log-uniform priors, respectively.

where Γ is the gamma function, and K_ν is the modified Bessel function of the second kind. $d_{ij} = \sqrt{(x_m - x_n)^2 + (y_m - y_n)^2}$ represents the distance between these two pixels. The parameter ρ defines the characteristic smoothing scale, while ν governs the order of smoothing, ensuring that the pixelized model reconstruction is $\lceil \nu \rceil - 1$ times differentiable. The Exponential and Gaussian kernels used in [Vernardos & Koopmans \(2022\)](#) are special cases of the Matérn kernel, with $\nu = 0.5$ and $\nu = +\infty$, respectively. The Hessian ($\mathbf{R}_{\delta\psi}$) of the regularisation function, using the Matérn scheme, is defined as follows:

$$\mathbf{R}_{\delta\psi}(\lambda^{\delta\psi}, \rho, \nu) = \lambda^{\delta\psi} \mathbf{C}_{\delta\psi}^{-1}(\rho, \nu), \quad (20)$$

where $\mathbf{C}_{\delta\psi}$ is the covariance matrix with elements C_{ij} , and $\lambda^{\delta\psi}$ controls the overall strength of the regularisation. Bilinear interpolation is employed to interpret the pixelized lensing potential perturbations as a continuous field.

3 OVERVIEW OF STRONG LENSING DATA

We present an overview of the galaxy-galaxy strong lensing data used in this work. Section 3.1 outlines the image simulation settings for the mock lenses employed to demonstrate the capability of our method. In Section 3.2, we describe the HST observational data for the lens system SLACS0946+1006, which is re-analysed using our new potential correction code.

3.1 Mock Lensing Data

We generate a set of mock lensing images that incorporate various perturbation signals in the lensing potential to evaluate the effectiveness of the potential correction code developed in this study. To control for variables, we ensure that the mass of the main lens galaxy

and the source light remain consistent across all mock datasets, allowing only the perturbation signals in the lensing potential to vary. The mass distribution of the main lens galaxy is described by the elliptical power-law model ([Tessore & Metcalf 2015](#)), whose dimensionless surface mass density (i.e. convergence) profile is expressed as

$$\kappa(r) = \frac{3 - \gamma}{2} \left(\frac{\theta_E}{r} \right)^{\gamma-1}. \quad (21)$$

Here, $1 < \gamma < 3$ represents the slope of the deprojected volume density, θ_E denotes the Einstein radius, and r signifies the elliptical radial distance from the centre of the lens galaxy. The elliptical radial distance is defined as $r(x, y) = \sqrt{x^2 q + \frac{y^2}{q}}$, where q is the minor-to-major axis ratio, and ϕ is the position angle measured counterclockwise from the positive x-axis. The parameters q and ϕ are reparametrized into two ellipticity components, ϵ_1 and ϵ_2 , defined as

$$\epsilon_1 = \frac{1 - q}{1 + q} \sin 2\phi, \quad \epsilon_2 = \frac{1 - q}{1 + q} \cos 2\phi, \quad (22)$$

to facilitate more efficient model fitting ([Birrer et al. 2015](#)). The surface brightness distribution of the source is characterised by the superposition of two elliptical Gaussian models, reflecting the irregular morphology observed in real lenses (e.g. [Bolton et al. 2008](#); [Etherington et al. 2022](#)).

Four popular types of lensing potential perturbation signals, frequently used to evaluate the performance of free-form models (e.g. [Vernardos & Koopmans 2022](#); [Galan et al. 2022](#); [Biggio et al. 2023](#)), are employed to construct mock datasets and test our potential correction code. These include: a spherical Navarro-Frenk-White (NFW) profile representing subhalos within the lens galaxy ([Navarro et al. 1996](#)), an external shear field caused by neighbouring galaxies or the cosmic large-scale structure ([Keeton et al. 1997](#)), a multipole perturbation induced by galaxy interactions such as mergers ([Hao et al.](#)

Main Lens's Mass Distribution		
Elliptical Power Law	Centre (y, x) ["]	(0.0, 0.0)
	Axis Ratio (q)	0.9
	Position Angle (ϕ) [°]	0.0
	Einstein Radius (θ_E) ["]	1.8
	Density Slope (γ)	2.0
	Redshift	0.2
Source Brightness Distribution		
Two Elliptical Gaussian Models	Centre (y, x) ["]	(0.0, 0.0)
		(0.0, 0.4)
	Axis Ratio	0.6
		0.4
	Position Angle [°]	45.0
		135.0
	Standard Deviation (σ) ["]	0.15
		0.1
	Intensity [$e^{-1} s^{-1}$]	5.0
		3.0
	Redshift	0.6
Lensing Potential Perturbations		
Spherical NFW Subhalo	Centre (y, x) ["]	(1.81, 0.0)
	Mass (M_{200}) [$M_\odot$]	10^{10}
	Concentration (c)	11.60
	Redshift	0.2
Shear	Magnitude (γ^{ext})	0.01
	Orientation (ϕ^{ext}) [°]	45.0
Multipole	Order (m)	4
	Normalization (a_m)	0.0569
	Orientation (ϕ_m) [°]	0.0
Gaussian Random Field	Normalization ($\log[A_{\text{GRF}}]$)	$-6.0 \left(-4.35^{0.03}_{-0.03} \right)$
	Power Spectrum Slope (β_{GRF})	$6.0 \left(2.72^{0.05}_{-0.05} \right)$

Table 2. Simulation setup for generating mock lensing images. The main lens mass and source light properties were chosen to match typical values from the SLACS sample. Mass perturbations were set to astrophysically plausible values. For the Gaussian random field, the values on the left correspond to the ground-truth power spectrum properties used to generate the lensing potential perturbation across the field of view. In contrast, the values in parentheses represent the power spectrum properties computed within the annular unmasked region where the lens modelling is performed. These properties were derived by fitting a power-law model to the one-dimensional power spectrum of lens potential perturbations calculated within this region, with superscripts and subscripts indicating the 68% confidence interval.

2006), and a Gaussian Random Field (GRF), which is expected to capture the collective lensing effects of numerous line-of-sight halos (Chatterjee & Koopmans 2018; Bayer et al. 2023). The NFW model has a volume mass density profile of

$$\rho = \frac{\rho_s}{(r/r_s)(1 + r/r_s)^2}, \quad (23)$$

where ρ_s is the normalization factor and r_s is the scale radius. ρ_s and r_s are typically reparametrized into the halo mass M_{200} and concentration $c = r_{200}/r_s$, where M_{200} is the mass enclosed within the radius r_{200} , within which the average density is 200 times the critical density of the Universe. Here, we assume that NFW halos follow the mass-concentration relation of Ludlow et al. (2016). Thus, the NFW model is solely quantified by M_{200} . The external shear is parameterized as two elliptical components ($\gamma_1^{\text{ext}}, \gamma_2^{\text{ext}}$), whose shear magnitude

γ^{ext} and orientation ϕ^{ext} measured counter-clockwise from the north are given by

$$\gamma^{\text{ext}} = \sqrt{(\gamma_1^{\text{ext}})^2 + (\gamma_2^{\text{ext}})^2}, \quad \tan 2\phi^{\text{ext}} = \frac{\gamma_2^{\text{ext}}}{\gamma_1^{\text{ext}}}. \quad (24)$$

The lensing potential of the shear model is usually expressed in polar coordinates (θ, φ) by

$$\psi^{\text{ext}}(\theta, \varphi) = -\frac{1}{2}\gamma^{\text{ext}}\theta^2 \cos 2(\varphi - \phi^{\text{ext}}). \quad (25)$$

The lensing potential of the multipole model (Xu et al. 2015; Van de Vyvere et al. 2022; Lange et al. 2024) is expressed as

$$\psi^{\text{mpole}}(\theta) = \frac{r}{1 - m^2} a_m \cos(m\phi - m\phi_m), \quad (26)$$

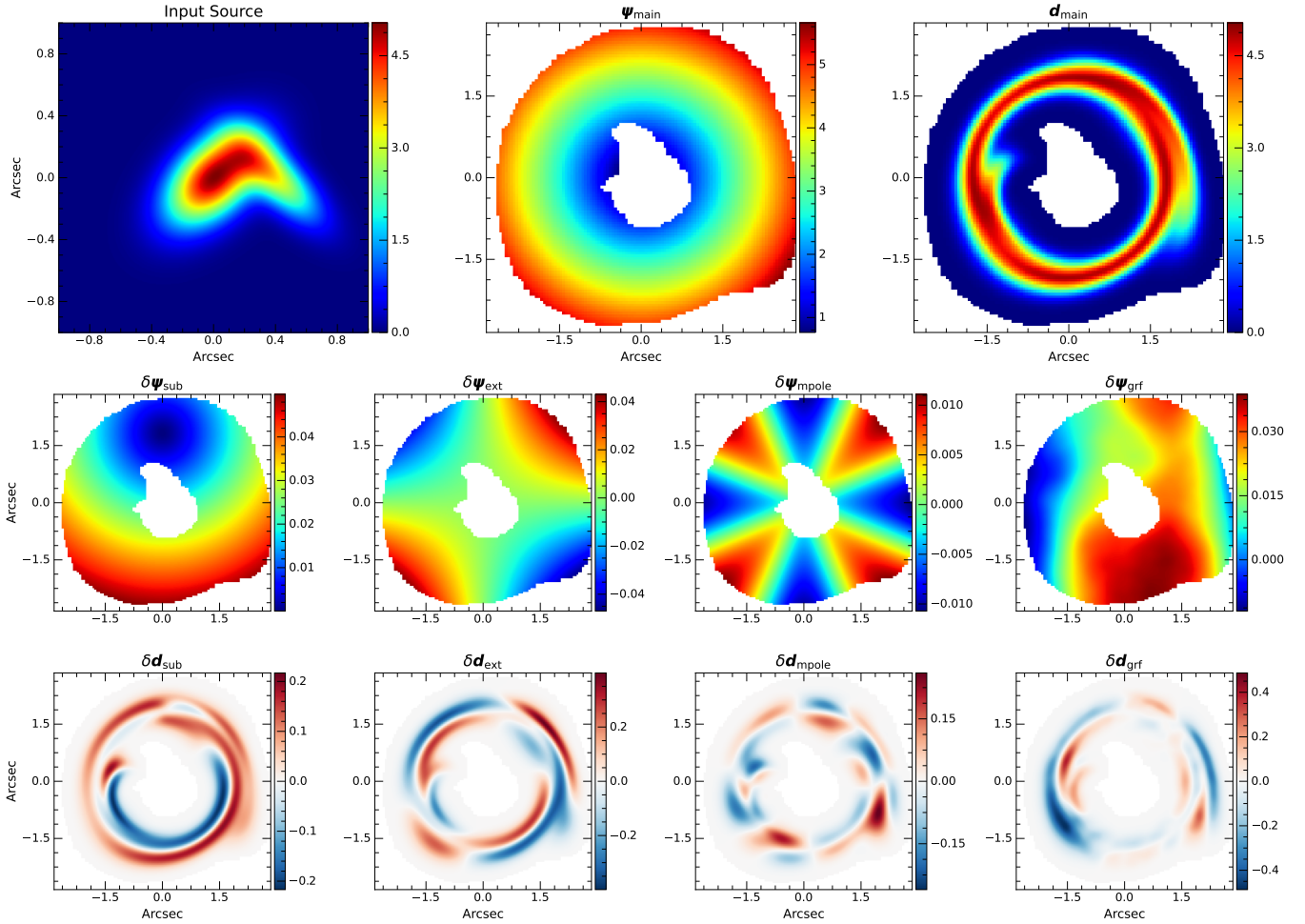


Figure 1. Mock datasets used in this work (see Table 2 for specific parameter values). The first row depicts the intrinsic source image, the lensing potential of the main lens, and the resulting lensed image. The second row shows the lensing potential maps of various perturbative signals: a $10^{10} \text{ M}_{\odot}$ NFW subhalo, an external shear field, a $m = 4$ multipole component, and a Gaussian random field. The third row presents the corresponding image perturbations induced by these lensing potential perturbations. Unless otherwise noted, lensing potentials are shown in units of arcsec^2 , and lensing images are in units of $\text{e}^- \text{ s}^{-1} \text{ pixel}^{-1}$.

where a_m represents the normalization factor, and m and ϕ_m denote the multipole order and orientation, respectively (c.f. Congdon & Keeton 2005; Taylor et al. 2017; Etherington et al. 2024; Paugnat & Gilman 2025). The characteristics of a GRF are defined by its isotropic power spectrum, which follows a power-law relation:

$$P(k) = A_{\text{GRF}} k^{-\beta_{\text{GRF}}}, \quad (27)$$

where A_{GRF} is the normalization factor, k is the wave number, and β_{GRF} is the slope. By specifying particular values for A_{GRF} and β_{GRF} , a random realisation of a GRF-like lensing potential field can be generated using

$$\psi^{\text{GRF}} = \mathcal{F}^{-1}(P(k), \epsilon), \quad (28)$$

where ϵ represents the random number seed, and $\mathcal{F}^{-1}$ denotes the inverse Fourier transform. We employ the powerbox software (Murray 2018) to perform the operations described in Equation 28.

The specific parameter settings used to generate the mock datasets are summarised in Table 2, with the corresponding lensing images presented in Figure 1. Since the potential correction method only works on an annular region encircling the lensed images, annular masks are automatically generated by identifying connected regions in the image where the pixel's signal-to-noise ratio (SNR) is greater than 3, then dilating these regions by 10 pixels. Our mock datasets

exhibit image quality comparable to that of the SLACS lens samples (Bolton et al. 2008), with a pixel size of $0.05''$ and a Gaussian PSF characterised by a σ value of $0.05''$. The sky background level is set to $0.1 \text{ e}^- \text{ s}^{-1}$, and the lensing images are subjected to exposures of 840 seconds. The maximum SNR (in the brightest pixel) of the lensed arc in our mock datasets is approximately 50, roughly corresponding to the brightest systems in the SLACS sample. For simplicity, all our mock lensing images omit the lens galaxy's light during the simulation. This simplification does not compromise the generality of this work, as the lens galaxy's light can be pre-subtracted before applying the potential correction method.

3.2 Lensing Data for SLACS0946+1006

SLACS0946+1006 is a compound lens system with a rare triple source-plane configuration, characterized by a lens redshift of $z_l = 0.222$ and source redshifts of $z_{s1} = 0.609$, $z_{s2} = 2.035$, and $z_{s3} = 5.975$ (Smith & Collett 2021). This system was initially pre-selected as a lens candidate from early-type galaxies in the SDSS survey, which exhibited abnormal emission lines potentially originating from high-redshift background sources (Bolton et al. 2006). The ACS camera on board the HST was subsequently employed to confirm

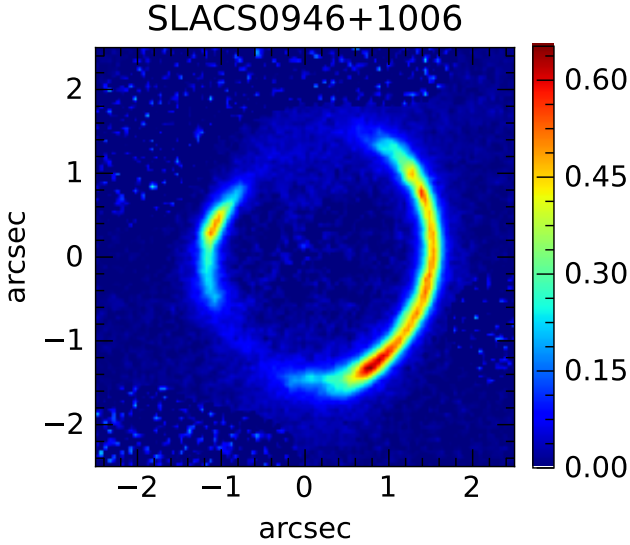


Figure 2. Lens-light subtracted image of SLACS0946+1006 in the HST F814W band, using the multiple Gaussian expansion model presented in He et al. (2024).

the strong lensing nature of this system. The SLACS0946+1006 data analysed in this work were obtained using the ACS F814W band and then drizzled to $0.05''\text{pixel}^{-1}$, following the reduction procedures described in Bolton et al. (2008). The PSF was determined using the TinyTim software. For simplicity, we masked the outer Einstein ring, using only the innermost Einstein ring for potential correction. The lens light was pre-subtracted using the model results from He et al. (2024), which employs the Multipole Gaussian Expansion (MGE) model to represent the lens light; combined with a pixelized source model, the lens and source light are fit simultaneously to account for the covariance between the two. Subtracting the lens light using MGE typically outperforms multiple Sersic (Sérsic 1963) fits in terms of image residuals. Figure 2 shows the lens-light-subtracted image of SLACS0946+1006.

4 RESULT

In Section 4.1, we assume perfect knowledge of both the source light and the image residuals induced by input mass perturbations to solve for the pixelised lensing potential perturbations from these image residuals, using the framework outlined in Section 2.2. Although this test disregards the potential degeneracy between the source light and the lens mass, which could affect the performance of a free-form lens model, it provides valuable insights into the viability of our free-form approach, particularly the effectiveness of regularisation schemes in recovering various forms of lensing potential perturbations. In Section 4.2, we evaluate the performance of our potential correction algorithm in a more realistic context by simultaneously modelling the pixelised lensing potential perturbations and source light through direct fitting of the lensing images (using the framework presented in Section 2.3), thereby accounting for their covariance. Since the macro model typically absorbs some of the image signals caused by perturbations, it introduces biases in the estimated main lens mass and source light. To assess the impact of these biases on the performance of the potential correction algorithm, we conduct two tests: one initialises the potential correction with the true input models of the main lens mass and source light (Section 4.2.1), while the

other uses the main lens mass and source light estimates derived from the macro model as the starting point (Section 4.2.2). Finally, in Section 4.3, we apply our method to the real HST lens system SLACS0946+1006 and compare our findings with previous studies.

4.1 Inverting Only the Potential Perturbation

We evaluate the performance of our proposed correction method using the mock datasets described in Section 3.1. We assume that the brightness distribution of the source and the image residuals induced by lensing potential perturbations (as illustrated in Figure 1) are perfectly known a priori. This information, along with the input mass distribution of the main lens, is then utilised to construct the vector $\delta\mathbf{d}$ and the matrix $\mathbf{D}_s$ in Equation 6. The framework presented in Section 2.2 is employed to solve for the lensing potential perturbations $\delta\psi$. Specifically, we apply the Matérn regularisation discussed in Section 2.4.2 to enforce smoothness constraints on $\delta\psi$. Given an arbitrary set of hyperparameters denoted by $\xi_{\delta\psi}$, the most probable solution $\delta\psi_{\text{mp}}$ can be derived using Equation 9. Assuming that the prior $P(\xi_{\delta\psi})$ for $\delta\psi$ is noninformative (see Table 1), the posterior distribution for the hyperparameters $P(\xi_{\delta\psi} | \delta\mathbf{d}, \mathbf{L}_{\delta\psi}, \mathbf{g}_{\delta\psi})$ is determined by sampling the likelihood $P(\delta\mathbf{d} | \mathbf{L}_{\delta\psi}, \xi_{\delta\psi}, \mathbf{g}_{\delta\psi})$, the specific form of which is given by Equation 10. The optimal hyperparameter values determined via Bayesian evidence are summarised in Table B1.

The model result for mock data with an input NFW subhalo perturber is shown in Figure 3. Our potential correction method successfully finds a solution for the lensing potential perturbations ($\delta\psi^M$) that produces a model image residual ($\delta\mathbf{d}^M$) closely matching the observed residual ($\delta\mathbf{d}^T$). Furthermore, $\delta\psi^M$ is highly consistent with the lensing potential map of the input perturber ($\delta\psi^T$).

To better visualise the structure of a localised subhalo, the convergence map is more informative than the lensing potential map. The top-left and top-right panels of Figure 4 compare the convergence maps of the input perturber and the one derived by the potential correction model, respectively. Our potential correction method with Matérn regularisation successfully reconstructs the mass distribution of the input NFW subhalo qualitatively. In contrast, using Exponential or Gaussian regularisation, as proposed in Vernardos & Koopmans (2022), leads to under- or over-smoothed solutions that fail to recover the input NFW subhalo, as shown in the bottom-left and bottom-right panels of Figure 4. This demonstrates the superiority of Matérn regularisation in reconstructing localised subhalo perturbations.

To enable a more quantitative assessment of the potential correction method, we fit a parametric model to the perturbative lensing potential map derived by the potential correction method for mock lenses with an NFW subhalo⁵, external shear, and an m_4 multipole perturber, and evaluate whether the fitting results agree with the true input values. For the Gaussian random field, we compare the power spectrum of the model-derived lensing potential perturbations with the true input values. As shown in Figure 5, the lensing potential perturbations derived by the potential correction method (second row) are consistent with the true input values (first row). A quantitative diagnostic (bottom row), conducted either via parametric model fitting or power spectrum comparison, demonstrates that our potential correction method can accurately recover the input perturbations. The

⁵ The NFW subhalo is a localised perturber. For this parametric model fitting, we only use the potential correction results in the central region (indicated by the black circle in Figure 4), as the outskirts are likely degraded by noise.

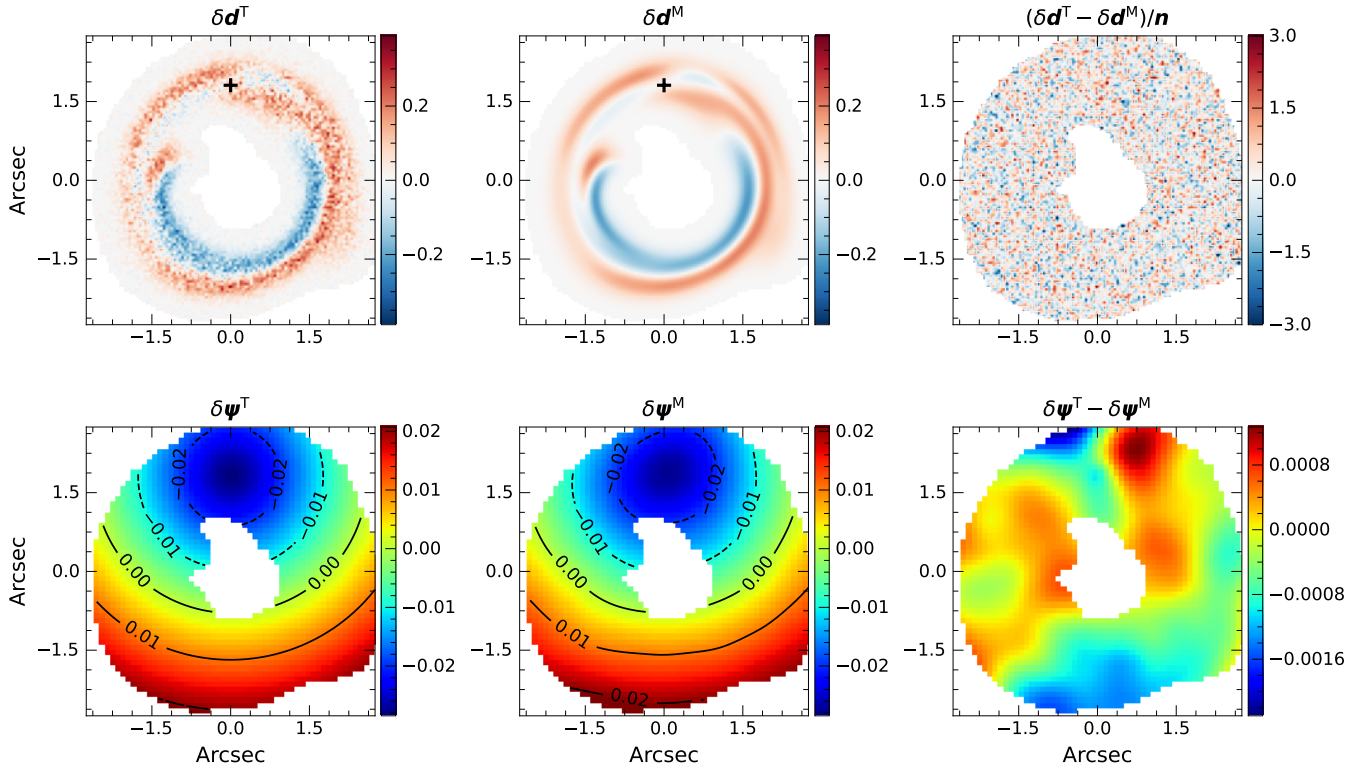


Figure 3. Reconstruction of the lensing potential perturbation induced by a $10^{10} M_{\odot}$ NFW subhalo using the Matérn regularisation scheme, assuming that the true source and the image residuals induced by the perturber are perfectly known. *Top-left:* Image residual induced by the $10^{10} M_{\odot}$ NFW subhalo. *Top-middle:* Image residual reconstructed using the potential correction model. *Top-right:* Difference between the top-left and top-middle panels, normalised by the noise map. *Bottom-left:* Lensing potential map of the input perturber. *Bottom-middle:* Lensing potential map of the model perturber, reconstructed using the potential correction model. *Bottom-right:* Difference between the bottom-left and bottom-middle panels. The ‘plus’ symbol in the top-left and top-middle panels marks the position of the input subhalo.

specific parametric values from this diagnostic are summarised⁶ in Table 3.

4.2 Simultaneous Inversion of Source and Potential Perturbations

4.2.1 Initialisation with True Lens and Source Models

We construct the matrices $\mathbf{L}_s$ and $\mathbf{L}_{\delta\psi}$, as defined in Equation (12), using the true input models for the main lens mass and source light. The semi-linear inversion framework described in Section 2.3 is then employed to solve for the pixelized source light (s) and lensing potential perturbations ($\delta\psi$). To regularise the solutions, `ConstantSplit` regularisation is applied to the pixelized source light, while Matérn regularisation is used for the lensing potential perturbations. The hyperparameters associated with the source and lensing potential perturbation regularisations are determined by maximising the Bayesian evidence, and the specific values of the latter are presented in Table B1.

As a demonstration, we present the results of simultaneously solving for the pixelized source and lensing potential perturbations for a

simulated lens with an input NFW perturber (Figure 6). Our method effectively recovers the lensed image, reducing model image residuals to the noise level. The reconstructed source closely resembles the input. Furthermore, we clearly observe a localised subhalo signal in the perturbative lensing potential map (manifested as a potential well) and in the corresponding convergence map (manifested as a positive mass clump).

Figure 7 compares the lensing potential perturbations derived from our method (second row) with the input truth (first row) for all four perturber types considered in this work: NFW subhalo, external shear, m_4 multipole, and Gaussian random field. The recovered perturbations agree closely with the input values. For a quantitative comparison, we fit each perturbation with a parametric model (for the NFW subhalo, external shear, and m_4 multipole) or calculate the power spectrum (for the Gaussian random field). We then verify whether the recovered perturbation properties, summarised in Table 3, are consistent with the input truth. We find that the recovered properties are generally statistically consistent with the input. This performance is comparable to tests that solve solely for the lensing potential perturbations, demonstrating that the lens-source degeneracy does not significantly degrade the recovery of various lensing potential perturbations by our method.

4.2.2 Initialisation with Macro Model Estimates

A procedure similar to that described in Section 4.2.1 is used to solve for the unknown source and lensing potential perturbations. However,

⁶ It should be noted that, although the power spectrum properties of the input Gaussian random field are recovered reasonably well using the potential correction method, the model and the input true values do not perfectly align within the statistical uncertainty, a phenomenon also observed in Vernardos & Koopmans (2022).

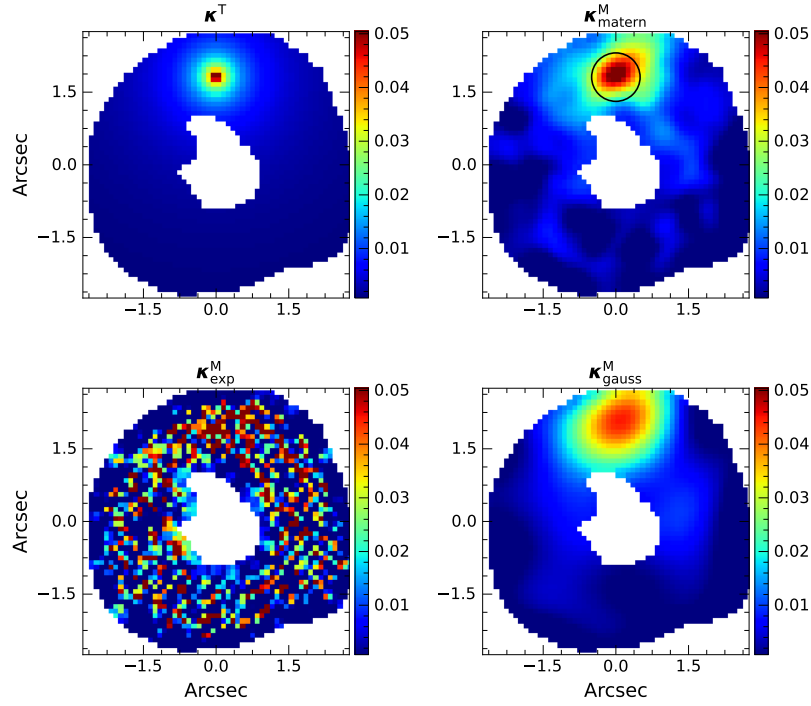


Figure 4. Reconstruction of the convergence perturbation induced by a $10^{10} M_{\odot}$ NFW subhalo using different regularisation schemes for the perturbative lensing potential. **Top-left:** Input convergence map of the $10^{10} M_{\odot}$ NFW subhalo. **Top-right:** Reconstructed convergence map using the potential correction method with Matérn regularisation. The black circle (radius $0.5''$) indicates the region where a parametric NFW model is fit to the reconstructed lensing potential to assess consistency with the input perturbation. **Bottom-left:** Same as top-right, but with Exponential regularisation. **Bottom-right:** Same as top-right, but with Gaussian regularisation. Exponential and Gaussian regularisations tend to produce under- or over-smoothed solutions, whereas the Matérn kernel yields a good reconstruction.

the matrices $\mathbf{L}_s$ and $\mathbf{L}_{\delta\psi}$ are constructed using the macro-model results. As our macro-model employs a pixelized source model, we derive the source gradient as follows to construct $\mathbf{L}_s$. For each Voronoi source cell, a square cross is centred on the cell, aligned with the source plane’s x and y axes. The side length of this square cross is equal to the square root of the cell’s area. We then evaluate the source brightness at the four endpoints of the cross using natural neighbour interpolation. The source brightness gradient for that Voronoi cell is then calculated using a finite difference approximation between the endpoints. Once the source brightness gradient has been calculated for every Voronoi cell, the gradient at any position on the source plane can be determined via further natural neighbour interpolation.

Figure 8 presents the potential correction results for the mock lens with an input NFW subhalo perturber. The model continues to fit the data exceptionally well, with the remaining image residuals reduced to the noise level. A localised perturber, representing the signal from the input subhalo, is clearly visible in the potential correction (first column panel in the bottom row) and the corresponding convergence correction map (second column panel in the bottom row). Since the macro model can absorb some perturbation signals from the input subhalo and provides a biased estimate of the input true main lens mass and source, the perturbation reconstructed by the potential correction method does not constitute a direct comparison with the input true perturbation. Therefore, we define a “pseudo-true perturbation” quantity, which is the perturbation derived from the potential correction method, denoted as $\delta\psi^M$, plus the lensing potential predicted by the macro model, ψ_{macro} , minus the input true main lens values, ψ_{main} . This “pseudo-true perturbation” quantity is more comparable to the input true values of the perturbations.

Figure 9 is analogous to Figure 7 and is employed to assess whether

starting from a biased estimation of the main lens mass and source degrades the performance of the potential correction for all four types of perturbations examined in this work. It is noteworthy that while the localised subhalo and m_4 multipole signal remain directly visible in the perturbative lensing potential map derived from the potential correction method (second row), the model perturbations of shear and the Gaussian random field do not resemble the input at all. This does not indicate a failure of the potential correction method, as the “pseudo-true perturbations” for shear and the Gaussian random field (third row), which provide a more appropriate comparison, remain in good agreement with the input. However, initiating the potential correction with inaccurate estimates of the main lens mass and source introduces small biases in the recovered perturbation fields when conducting quantitative diagnoses. As shown in Table 3, the shear orientation is offset by $\sim 2^\circ$ compared to the input truth, and the subhalo’s mass exhibits a tendency to be underestimated by ~ 0.5 dex.

4.3 Characterising the Subhalo in the Jackpot Lens via a Free-Form Approach

To apply the potential correction method to the Jackpot lens, we first generate an annular mask around the lensed arc by connecting pixels with a lensed image SNR greater than 3 and expanding the connected region by 5 pixels. Next, we use the inferred mass model derived by He et al. (2024), where the main lens’s mass is described by the EPL plus shear model, as the starting point to rerun a macro model that does not account for any perturbations. This refined macro model employs a pixelized source model with Kmeans pixelization

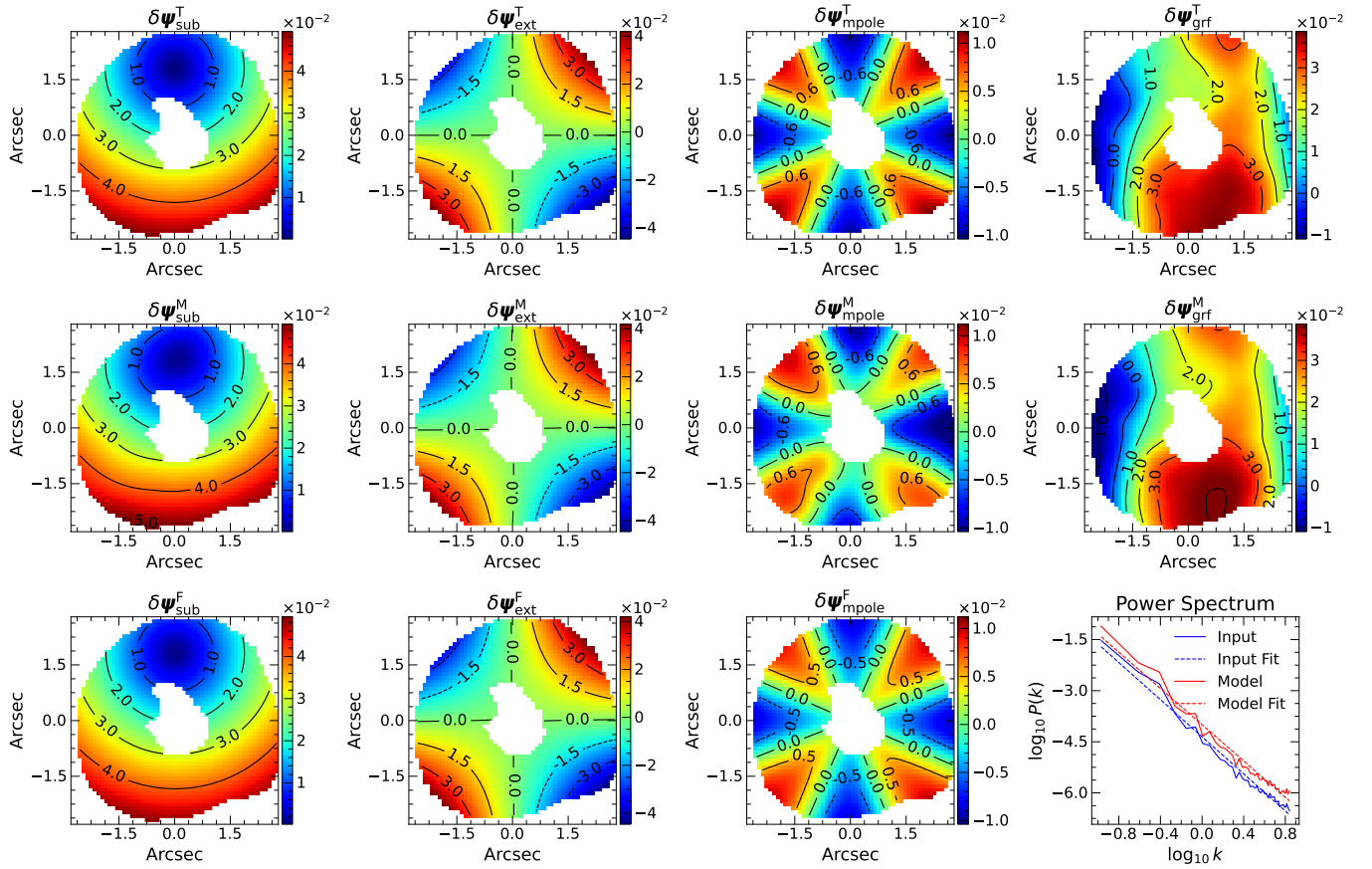


Figure 5. Comparison of reconstructed lensing potential perturbations induced by an NFW subhalo (left column), external shear (middle-left column), m_4 multipole (middle-right column), and a Gaussian random field (GRF) (right column) with the ground truth. The first row presents the input lensing potential maps of perturbations. The second row shows the perturbative lensing potential maps derived from the potential correction model. The third row presents a quantitative analysis: for the NFW subhalo, external shear, and m_4 multipole, parametric models are fitted to the lensing potential map in the second row, with point estimates reported in Table 3; for the GRF, the power spectra of the input (blue lines) and reconstructed (red lines) lensing potential maps are compared. Solid lines indicate power spectra calculated within the unmasked modelling region, while dashed lines represent power-law model fits.

and AdaptiveBrightnessSplit regularisation. The inferred mass model and the corresponding pixelized source reconstruction are then used to construct the $\mathbf{L}(\psi_p, s_p)$ matrix and solve for the unknown lensing potential perturbations, as detailed in Section 2.3. The hyperparameters associated with the source model, including Kmeans pixelization and AdaptiveBrightnessSplit regularisation, are fixed to the values obtained from the macro model when performing the potential correction. Allowing these hyperparameters to vary as free parameters is numerically unstable and renders the matrix equation 12 unsolvable.

Figure 10 presents the potential correction results for the Jackpot lens. Our model fits the observed lensing image exceptionally well, reducing the remaining image residuals to the noise level. A localised mass perturbation attributed to a subhalo is evident in the lensing potential perturbation map (manifested as a negative lensing potential well) and the corresponding convergence perturbation map (manifested as a positive mass clump). The central position of this localised mass perturber, defined as the pixel position with the maximum value in the convergence perturbation map, is $(y, x) = (-0.65, -1.05)$, consistent with the detection by Vegetti et al. (2010).

Minor et al. (2017) define a model-independent method to measure the “robust mass” of the subhalo, which is the aperture mass within the so-called robust radius R_{robust} . This radius can be calculated based on the geometric perturbations induced by the subhalo

on the main lens’s critical lines. For the subhalo in the Jackpot lens, $R_{\text{robust}} \approx 1$ kpc (Minor et al. 2021), as indicated by the black circle in Figure 10. This yields a robust mass of $3 \times 10^8 M_\odot$ based on our potential correction result. However, this value is an order of magnitude lower than the result reported by Minor et al. (2021), who explicitly used a truncated NFW model to describe the subhalo’s mass, deriving a robust mass of $\sim 2 - 3.7 \times 10^9 M_\odot$ ($> 95\%$ confidence level). We attribute this discrepancy to the limitations of our regularisation scheme, which, although it clearly performs better than the alternatives (as shown in Figure 4), still lacks the dynamic range needed to reconstruct highly compact perturbations. Specifically, we impose a constant regularisation strength for lensing potential perturbations across the entire modelling region. Although Bayesian evidence determines a relatively strong regularisation strength to suppress noise-like features in regions without significant perturbations, this approach simultaneously oversmooths the structure of highly compact perturbations. Notably, this limitation is analogous to the challenges faced when modelling clumpy sources with steep brightness gradients. In Section 5.1, we discuss how adaptive regularisation for lensing potential perturbations could mitigate these systematics.

The properties of the localised perturber in the Jackpot lens have been systematically studied using parametric approaches. These studies assume the perturber is either a subhalo with various mass profiles, such as NFW, truncated NFW, NFW following the mass-

Spherical NFW Subhalo				
Parameter Names	$\delta\psi$ -Only (§ 2.2)	$(\delta\psi, s)^{\text{Truth}}$ (§ 4.2.1)	$(\delta\psi, s)^{\text{Macro}}$ (§ 4.2.2)	Ground Truths
Centre (y, x) ["]	$(1.8190^{0.1774}_{-0.1798}, 0.0154^{0.1703}_{-0.1797})$	$(1.7714^{0.1716}_{-0.1668}, 0.0422^{0.1656}_{-0.1699})$	$(1.7634^{0.0480}_{-0.0498}, 0.0455^{0.0473}_{-0.0474})$	(1.81, 0.0)
Mass ($\log_{10}[M_{200}/M_{\odot}]$)	$9.9665^{0.7155}_{-0.6685}$	$9.9151^{0.7237}_{-0.6488}$	$9.5192^{0.8085}_{-0.3837}$	10.0
Shear				
Magnitude (γ^{ext})	$0.0101^{0.0008}_{-0.0008}$	$0.0100^{0.0008}_{-0.0008}$	$0.0110^{0.0001}_{-0.0001}$	0.01
Orientation (ϕ^{ext}) [°]	$45.6705^{2.4147}_{-2.3474}$	$44.5560^{2.4775}_{-2.3296}$	$47.3874^{0.3598}_{-0.3756}$	45
Multipole				
Normalization (a_m)	$0.0507^{0.0006}_{-0.0007}$	$0.0527^{0.0012}_{-0.0011}$	$0.0478^{0.0010}_{-0.0011}$	0.0569
Orientation (ϕ_m) [°]	$-0.0208^{3.1411}_{-3.1396}$	$-0.0081^{4.7032}_{-3.1458}$	$-0.0251^{3.1449}_{-3.1467}$	0.0
Gaussian Random Field				
Normalization ($\log[A_{\text{GRF}}]$)	$-4.00^{0.04}_{0.04}$	$-4.48^{0.03}_{-0.03}$	$-4.17^{0.04}_{-0.04}$	$-4.35^{0.03}_{-0.03}$
Power Spectrum Slope (β_{GRF})	$2.66^{0.06}_{-0.06}$	$2.70^{0.05}_{-0.05}$	$2.56^{0.07}_{0.07}$	$2.72^{0.05}_{-0.05}$

Table 3. Quantitative comparison of reconstructed and input lensing potential perturbations. For subhalo, shear, and multipole models, parametric models are fitted to the reconstructed perturbations to assess agreement with the input. For the Gaussian random field, the power spectrum of the perturbations is compared. The labels $\delta\psi$ -Only, $(\delta\psi, s)^{\text{Truth}}$, and $(\delta\psi, s)^{\text{Macro}}$ denote inversions of only the lensing potential perturbations (§ 2.2), inversions of both source and lensing potential perturbations starting from the exact true main lens mass and source light distributions (§ 4.2.1), and inversions of both source and lensing potential perturbations starting from the macro model fit (§ 4.2.2), respectively. Superscripts and subscripts represent the 68% credible interval.

concentration relation, pseudo-Jaffe, or power-law models (Vegetti et al. 2010; Minor et al. 2021; Nightingale et al. 2024; Ballard et al. 2024; Minor 2024; Despali et al. 2024), or line-of-sight halos (LOS-halos) (Enzi et al. 2024). In the left and middle panels of Figure 11, the convergence maps of the subhalo in the Jackpot lens, modelled by Despali et al. (2024) and Nightingale et al. (2024), are compared, with the lensed arc shown as contours. The subhalo detected by Despali et al. (2024) exhibits a more concentrated structure and is positioned closer to the lensed arc compared to the modelling result of Nightingale et al. (2024). This difference highlights how the assumed parametric functional form can influence subhalo measurements. We revisit this problem using a fully free-form approach that does not rely on any predefined functional form for the mass perturbations unaccounted for by the macro mass model. As shown in the right panel of Figure 11, the mass perturber derived using our potential correction method exhibits a convergence distribution that strongly aligns with the parametric model result of Despali et al. (2024), rather than that of Nightingale et al. (2024), in terms of both position and compactness. This result represents the first application of a fully free-form model to reveal the compact nature of the subhalo in the Jackpot lens.

While the flexible nature of the free-form model allows it to avoid many systematics present in parametric approaches, this flexibility can also make the performance of free-form models less stable than parametric ones. Therefore, we conduct several tests to evaluate the robustness of our potential correction results for the Jackpot lens. First, we applied a larger annular mask to the Jackpot lens and performed the potential correction again, finding that the properties of the reconstructed subhalo remained unchanged. Second, the poten-

tial correction results reveal additional features in addition to the detected subhalo. For instance, a dipole-like structure appears near the detected subhalo (in the top-right region), and a negative convergence peak is observed near $(x, y) \approx (1.0, 1.0)$. To investigate the origin of these features, we employ a parametric approach analogous to that used by Despali et al. (2024) to model the Jackpot lens and simulate Jackpot-like mock lenses based on this model. Applying our potential correction method to these mock lenses reproduces the same additional features. Further analysis suggests that these arise from initiating the potential correction with a biased estimate of the main lens mass and source light provided by the macro model. Third, we simulate two Jackpot-like mock lenses with subhalo properties given by Despali et al. (2024) and Nightingale et al. (2024), respectively. After applying our potential correction method to these mock lenses, we observe a localised perturber in both cases. Moreover, the perturber exhibits a more compact structure in the mock lens generated using the model of Despali et al. (2024). This test demonstrates our potential correction method's ability to reveal the compactness of a localised perturber. A detailed description of the robustness tests conducted here is provided in Appendix C.

5 DISCUSSION

Compared to prior works, our potential correction code offers several key advantages:

- It simultaneously reconstructs the pixelized source and the lensing potential perturbation, thereby accounting for the covariance between them.

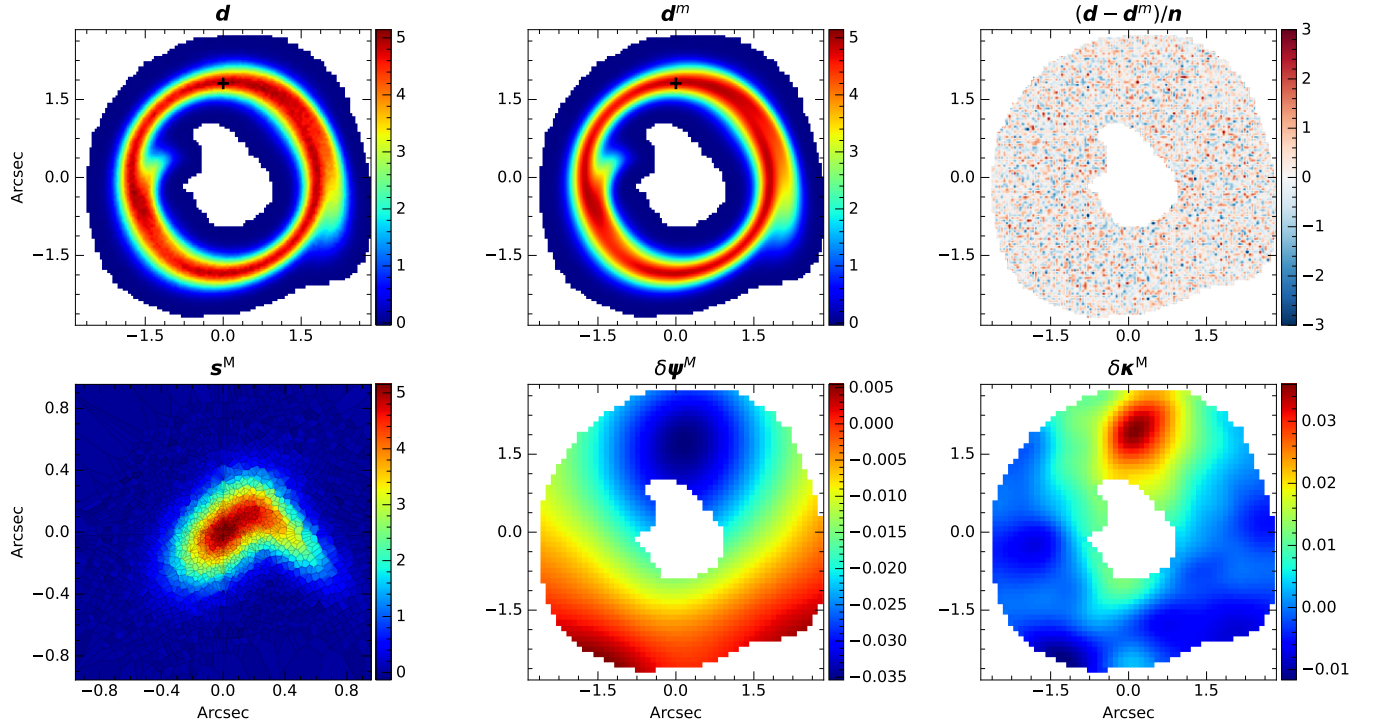


Figure 6. Results of simultaneous source and perturbative lensing potential modelling to directly fit the observed lensing image, using the input main lens mass and source light models as the starting point. From left to right: the first row displays the data ($\mathbf{d}$), model ($\mathbf{d}^m$), and normalised residual ($(\mathbf{d} - \mathbf{d}^m)/n$) images. The second row shows the reconstructed source ($\mathbf{s}^M$), the lensing potential correction ($\delta\psi^M$), and the corresponding convergence correction ($\delta\kappa^M$). The ‘plus’ symbol in the top-left and top-middle panels marks the position of the input subhalo.

- It objectively determines the regularisation parameters for the pixelized source and the lensing potential perturbation by optimising the Bayesian evidence.

- Its versatility allows for the recovery of both localised and extended lensing potential perturbations, facilitated by Matérn regularisation.

- The entire modelling procedure is automatic and requires no manual fine-tuning.

The efficacy of our programme has been demonstrated using both simulated and observational datasets, as presented in Section 4. Despite these impressive results, the current method has limitations, and there is scope for further improvement. These aspects are discussed in detail below.

5.1 Enhanced Regularisation Approaches for Lensing Potential Perturbations

When a pixelized free-form model is employed for lensing potential perturbations, strong regularisation is required to suppress noise-like signals in regions where perturbations are absent. Conversely, in regions where localised perturbations with high spatial gradients emerge, the regularisation strength should be reduced to allow more perturbation structures to be determined from the data rather than being overly smoothed by the prior (i.e., the regularisation). Assuming a constant regularisation strength across the entire modelling region fails to satisfy the differing requirements of regions with and without perturbations simultaneously. The optimal regularisation strength, as determined by Bayesian evidence, must serve as a compromise: it will be too weak for regions lacking perturbations and too strong for regions with highly localised perturbations. Consequently, a con-

stant regularisation scheme tends to over-smooth the structures of highly localised perturbations, such as subhalos. To fully address these systematic issues, adaptive regularisation schemes that allow for a broader dynamic range in lensing potential perturbations can be employed. Specifically, when localised perturbations are identified in the potential correction results obtained through a constant regularisation scheme, this information can be utilised as a prior. The regularisation strength can then be adjusted accordingly, increasing it in regions lacking significant mass signals and decreasing it in areas where localised mass structures are present.

Our potential correction framework is based on the quadratic form of the regularisation function, which allows the unknown perturbations in the lensing potential to be solved linearly. It has been demonstrated that quadratic regularisation with the Matérn kernel can effectively recover various types of lensing potential perturbations, whether extended or localised. Despite these impressive results, we observe certain artefacts in the potential correction results, such as random negative-convergence features, which have also been noted in previous works (Vegetti et al. 2010). These artefacts stem from the inherent nature of quadratic regularisation (also known as L2 regularisation or ridge regression), which promotes smoothness by shrinking all feature values (specifically, the lensing potential perturbations at each pixel) without setting any to zero (Hoerl & Kennard 1970a,b). To improve the potential correction method, an alternative approach to imposing smoothness priors may involve setting the lensing potential perturbation values at each pixel to zero, unless the data indicate otherwise. This requirement aligns with the purpose of L1 regularisation (or LASSO; Tibshirani 1996). However, utilising L1 regularisation disrupts the linear inversion framework, preventing us from using equation 9 to solve for the lensing potential perturbation. Instead, we must determine the lensing potential pertur-

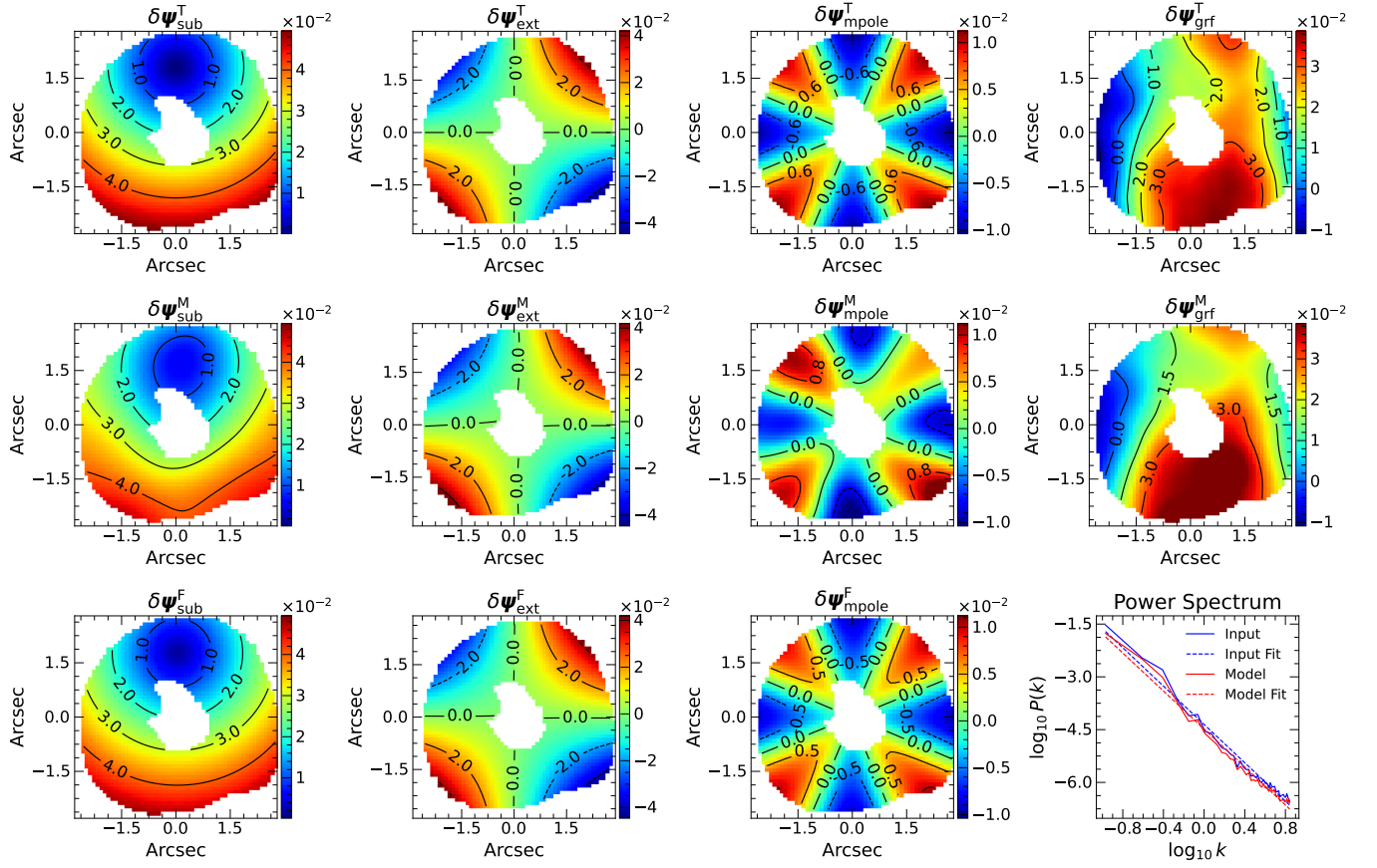


Figure 7. Analogous to Figure 5, but for the simultaneous modelling of the source and lensing potential perturbations, assuming perfect knowledge of the true main lens mass and source.

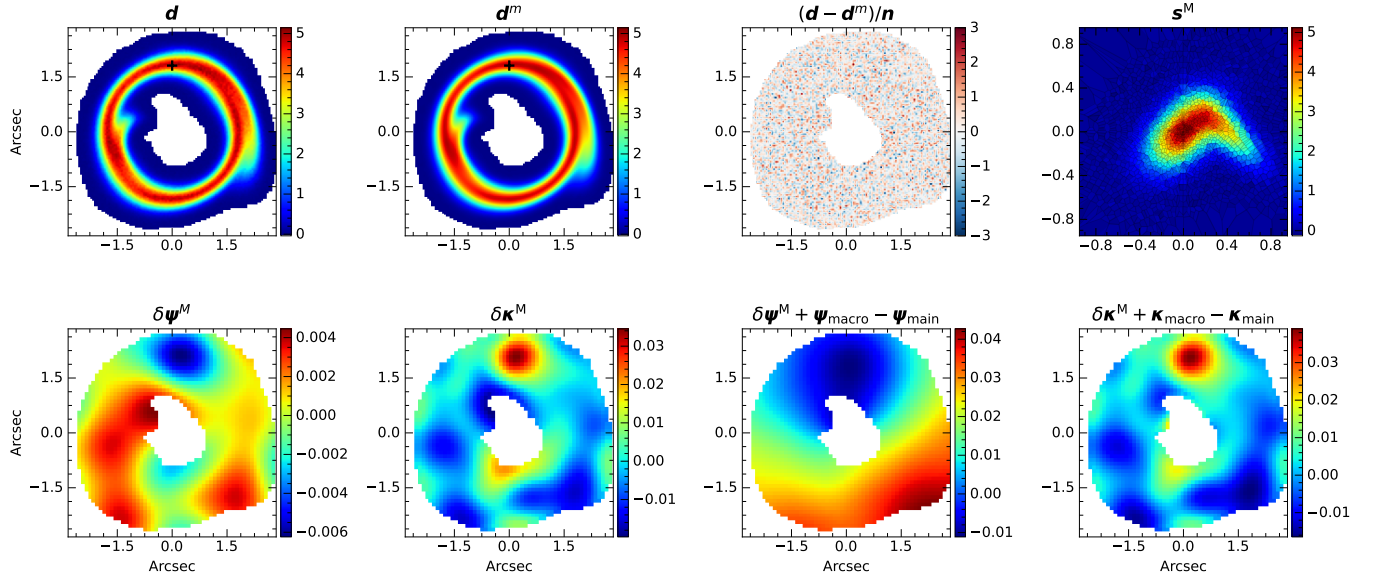


Figure 8. This figure replicates the structure of Figure 6, but initiates the potential correction from the main lens mass and source light models derived by the macro model. As the macro model yields a biased estimate of the main lens mass, we also present a “pseudo-true” lensing potential (convergence) perturbation map, defined as the model-derived perturbative lensing potential (convergence) plus the one given by the macro model minus the input true main lens values: $\delta\psi^M + \psi_{\text{macro}} - \psi_{\text{main}}$ ($\delta\kappa^M + \kappa_{\text{macro}} - \kappa_{\text{main}}$).

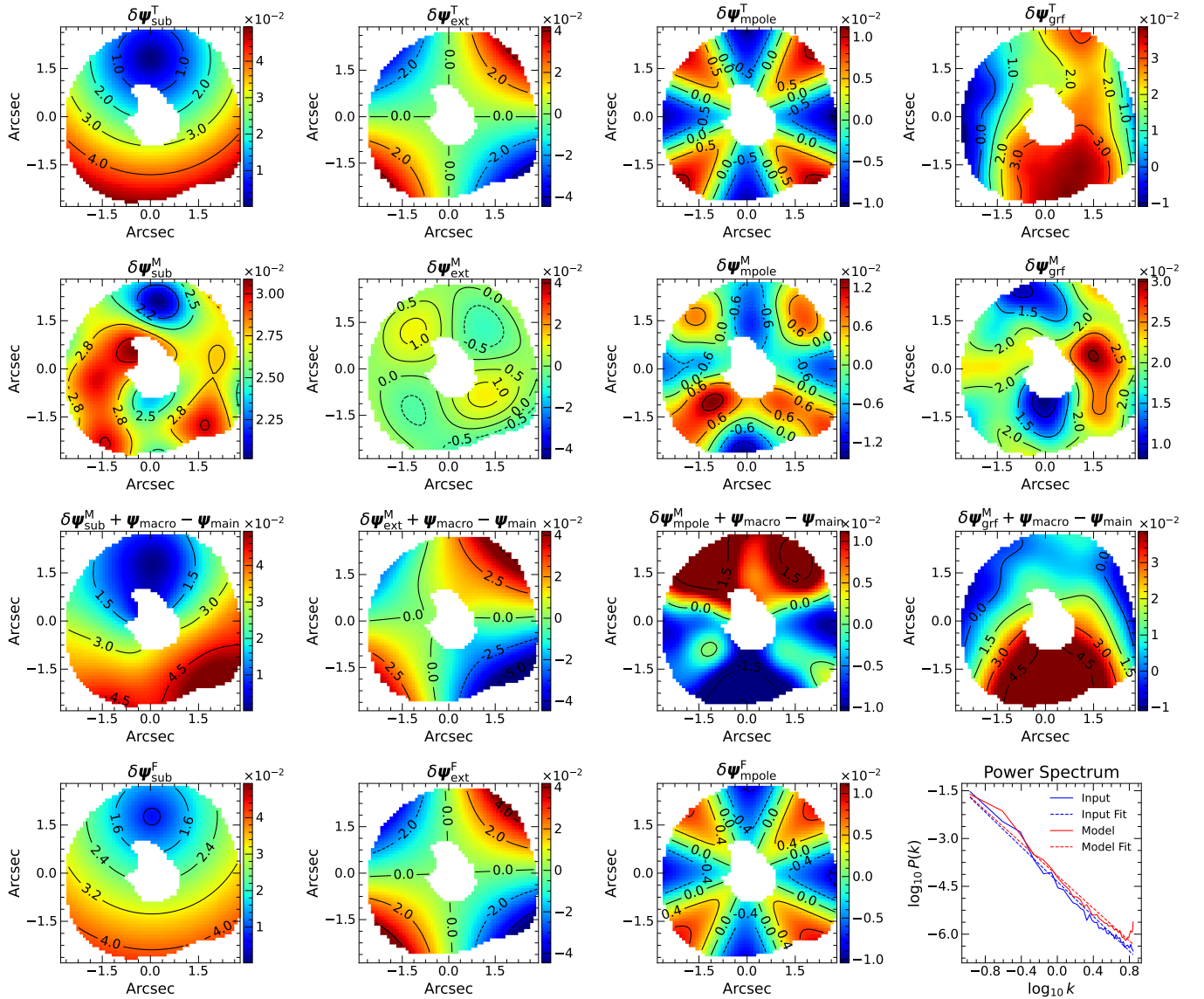


Figure 9. Analogous to Figure 7, but starting the potential correction with a biased estimate of the main lens mass and source light derived from the macro model. The third row displays the “pseudo-true” lensing potential perturbation maps, calculated as the sum of the model-derived perturbative lensing potential (convergence) and the macro model potential, minus the true input main lens potential ($\delta\psi^M + \psi_{\text{macro}} - \psi_{\text{main}}$). The quantitative diagnostics in the fourth column, including parametric model fitting and power spectrum calculation, are also based on these “pseudo-true” lensing potential perturbation maps.

bation values by directly optimising the penalty function illustrated in equation A7. This approach introduces a substantial number of nonlinear modelling parameters (several thousand), which are challenging for optimisers that do not leverage gradient information to navigate parameter space. Since our current code lacks support for auto-differentiation, we are unable to compute the gradient of the likelihood with respect to the modelling parameters, rendering our code incompatible with L1 regularisation at this time. Several open-source lens modelling software packages have incorporated auto-differentiation (Gu et al. 2022; Galan et al. 2022; Cao et al. 2025), and PyAutoLens plans to integrate this functionality soon. We plan to incorporate L1 regularisation (and potentially more general forms) into our code in future work.

5.2 Quantifying Uncertainties in Lensing Potential Perturbations

The artefacts in our current potential correction results are unavoidable due to the limitations of quadratic regularisation schemes. However, if we can quantify the uncertainties in the model-derived lensing potential perturbations and demonstrate that the signal in these artefact regions is highly uncertain, these artefacts will not impede the interpretation of the potential correction results. Given a set of hyperparameters for the source and lensing potential perturbation regularisations, which are best determined by maximizing the Bayesian evidence, we can estimate the uncertainties of the pixelized source and lensing potential perturbations using the covariance matrix, the mathematical form of which is provided in Equation A8⁷. This co-

⁷ Replace $\mathbf{L}_{\delta\psi}$ with $\mathbf{L}$.

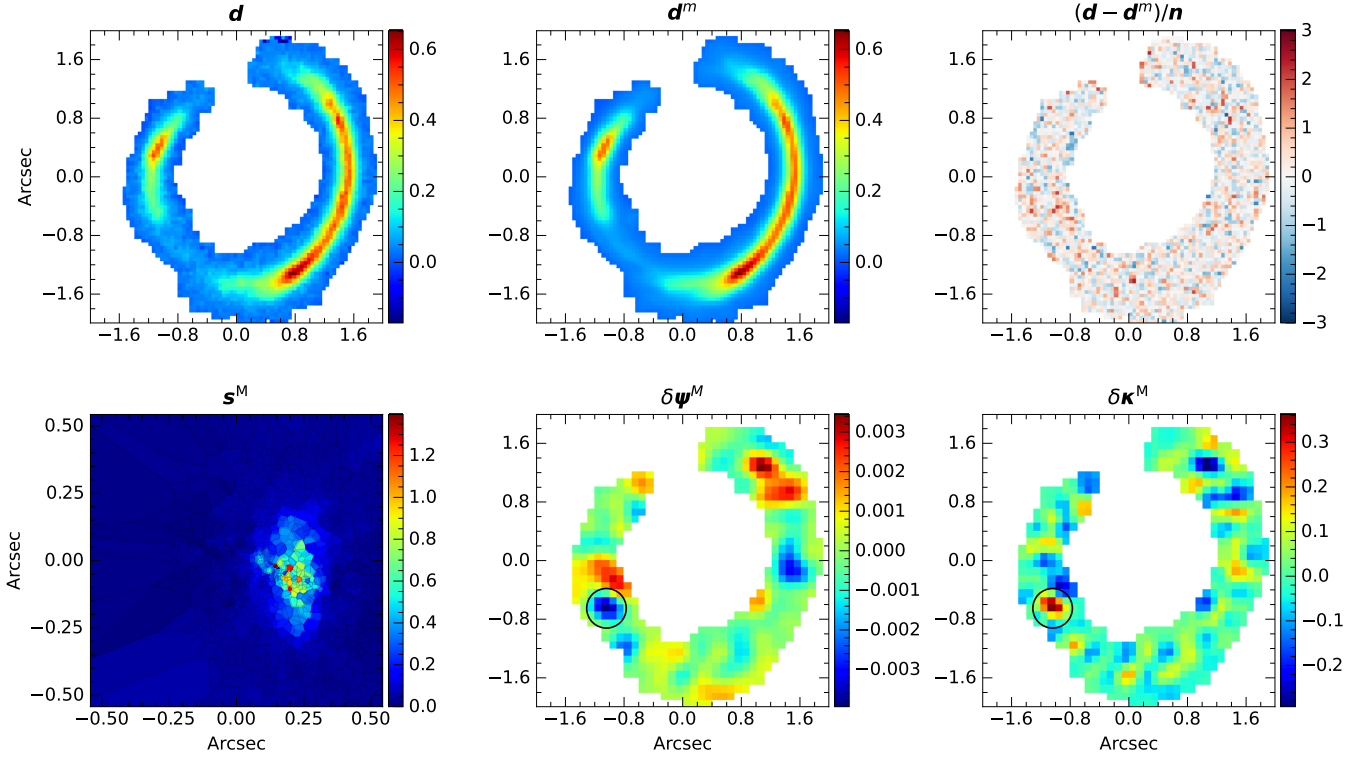


Figure 10. Potential corrections for SLACS0946+1006, analogous to Figure 8. In the bottom middle and right panels, black circles are centred on the peak of the localised convergence perturbation, with a radius of $0.27''$, corresponding to a physical scale of 1 kpc at the lens redshift.

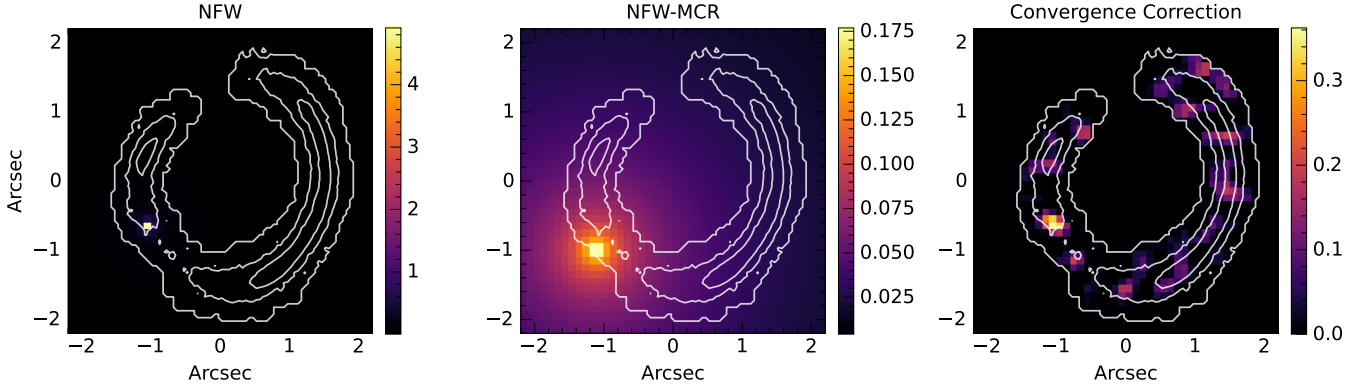


Figure 11. Convergence maps of the SLACS0946+1006 subhalo from different studies. **Left:** A $\sim 10^{10} M_\odot$ NFW subhalo with a high concentration of ~ 200 , centered at $(-1.040, -0.651)$ (Despali et al. 2024). **Middle:** A $\sim 10^{11.5} M_\odot$ NFW subhalo following the standard mass-concentration relation, located at $(-1.00, -1.10)$ (Nightingale et al. 2024). **Right:** The perturbative convergence map resulting from our potential correction. Black contours indicate the lensed image morphology.

variance matrix contains numerous non-zero off-diagonal elements, indicating a high degree of correlation between the pixelized source and lensing potential perturbation reconstructions. Since our primary interest lies in determining whether a mass perturbation structure is genuine, we aim to quantify the uncertainties of pixelized mass perturbations and define an SNR-like quantity to evaluate the fidelity of the perturbation signal. However, we currently lack a robust method to translate the covariance matrix of the lensing potential perturbations into uncertainties for the mass perturbations. Consequently, we resort to Monte Carlo simulations to estimate the uncertainties of pixelized mass perturbations, as detailed in Appendix C.

5.3 Incorporating Lens Light Modelling within Potential Correction Algorithms

The framework presented in Section 2.3 implicitly assumes that the light contribution from the lens galaxy has been perfectly subtracted from the lensing image prior to applying the potential correction method. However, pre-subtracting lens light limits our ability to fully account for the covariance between lens light and lensed source light. It has been demonstrated that lensing potential perturbations can be partially degenerate with the lens light signal (Nightingale et al. 2024). Therefore, a more effective approach to managing lens light is to integrate its modelling within the potential correction framework.

This can be achieved using the mathematical framework presented in [He et al. \(2024\)](#); [Ritondale et al. \(2019\)](#).

Suppose we utilise N_l elliptical Gaussian models that share a common centre and ellipticity, but have different standard deviations equally spaced in a logarithmic radial range (e.g. from 10^{-2} to $3''$) to represent the lens light. The image $\mathbf{d}^l$ formed by these Gaussian models can be expressed in the following matrix equation:

$$\mathbf{d}^l = \mathbf{B}\mathbf{L}_l(\xi_l)\mathbf{l}. \quad (29)$$

Here, the matrix $\mathbf{L}_l(\xi_l)$ has dimensions $[N_d, N_l]$, with each column representing the model image formed by each Gaussian model with a "unit" normalization value; ξ_l denotes the nonlinear parameters corresponding to the centre and ellipticities of the Gaussian models in this instance. The column vector $\mathbf{l}$ contains N_l elements, representing the normalisation factors for each Gaussian model. Equation 12 can now be extended to include lens light modelling as follows:

$$\begin{aligned} \mathbf{d} &= \mathbf{B}\mathbf{L}_s(\psi_p)\mathbf{s} + \mathbf{B}\mathbf{L}_{\delta\psi}(s_p)\delta\psi + \mathbf{B}\mathbf{L}_l(\xi_l)\mathbf{l} + \mathbf{n} \\ &= \mathbf{B}\mathbf{L}_{\text{all}}(\psi_p, s_p, \xi_l)\mathbf{r}_{\text{all}} + \mathbf{n}. \end{aligned} \quad (30)$$

Here, the block matrices

$$\mathbf{L}_{\text{all}}(\psi_p, s_p, \xi_l) \equiv \left(\mathbf{L}_s(\psi_p) \mid \mathbf{L}_{\delta\psi}(s_p) \mid \mathbf{L}_l(\xi_l) \right), \quad (31)$$

and

$$\mathbf{r}_{\text{all}} \equiv \begin{pmatrix} s \\ \delta\psi \\ \mathbf{l} \end{pmatrix}. \quad (32)$$

The Hessian of the regularisation function for the vector $\mathbf{r}_{\text{all}}$ can be expressed as:

$$\mathbf{R}_{r_{\text{all}}} = \begin{pmatrix} \mathbf{R}_s & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_{\delta\psi} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \epsilon \mathbf{I}_l \end{pmatrix}, \quad (33)$$

where $\mathbf{I}_l$ is an identity matrix of dimensions $[N_l, N_l]$, and ϵ is a small constant (e.g., 10^{-5}). Equations 12 and 30 retain the same mathematical form; thus, the linear inversion framework outlined in Section 2.3 remains applicable for potential corrections that incorporate lens light modelling.

5.4 Concurrent Modelling of Main Lens and Lensing Potential Perturbations

In Section 4.2.2, we demonstrated that the perturbations derived using the potential correction method do not generally match the input ground truth, as the macro model can vary to absorb some of the mass perturbations. To enable the potential correction method to recover the underlying true perturbations, a natural approach would be to simultaneously fit the main lens mass and the lensing potential perturbations (e.g. [Vernardos & Koopmans 2022](#)). Although we cannot currently perform this simultaneous fitting due to a fixed main lens mass model being hard-coded in our programme, we observe that starting the potential correction from the input true main lens mass model yields higher Bayesian evidence than starting from the macro model's estimate. This suggests that simultaneous fitting of the main lens mass and lensing potential perturbations might be feasible. We plan to explore this possibility in future work.

6 SUMMARY AND CONCLUSION

We introduce a new free-form lens modelling tool built upon the open-source software PyAutoLens. Our approach models perturbations to the smooth lensing potential derived by a macro model,

as free parameters defined on a pixelized grid. By applying only minimal smoothing priors to these pixelized perturbations, we mitigate many systematic errors commonly encountered in parametric lens modelling arising from the strong priors imposed by predefined functional forms. Our key results are:

- Our code can recover various forms of perturbation signals, both extended and localised, in an automated manner. It avoids the fine-tuning issues that plague iterative potential correction methods by employing the Matérn regularisation kernel and objectively determining the hyperparameters for lensing potential perturbations and source regularisations through Bayesian evidence optimisation. Although initially developed for substructure detection, our code's ability to recover various forms of mass perturbation can also benefit other applications requiring the assessment of biases introduced by assumed parametric functional forms, such as time-delay cosmography (e.g. [Suyu et al. 2009](#); [Kochanek 2021](#)).
- Properties of the lensing potential perturbations are recovered without significant bias in mock tests. Objectively determining the regularisation parameters enhances the capability of our potential correction method to quantitatively reproduce the perturbation field.
- Degeneracies between the lens and source do not significantly impair the ability of our correction method to recover the properties of lensing potential perturbations, even when starting from a biased initial estimate of the lens and source, although there is a slight reduction in accuracy.
- Our analysis reveals a localised mass concentration within the Jackpot strong-lensing system. Its position and compact morphology are consistent with the subhalo detection obtained from parametric modelling by [Vegetti et al. \(2010\)](#), but contrast with the results of [Nightingale et al. \(2024\)](#). Because our method is entirely free-form, it circumvents several systematic uncertainties inherent to parametric lens modelling, particularly the degeneracy between the subhalo and the main lens mass distributions. Consequently, our findings provide compelling evidence that the subhalo in the Jackpot strong lens is over-concentrated, potentially challenging the standard cosmological model (e.g. [Minor et al. 2021](#)). These results may necessitate alternative self-interacting dark matter models to explain the anomalous subhalo properties observed in the Jackpot lens (e.g. [Li et al. 2025](#)).

Our current implementation applies constant regularisation across the entire modelling area. While this constant regularisation effectively suppresses noise in regions without obvious perturbations, it overly smooths the structure of highly compact mass clumps, particularly within the central cuspy region. Consequently, our method tends to underestimate the mass of highly compact mass clumps, such as a $10^{10}M_\odot$ NFW subhalo with a high concentration of ~ 200 . Future work will involve implementing adaptive regularisation to overcome this limitation. We anticipate incorporating an auto-differentiation framework into our potential correction method. This enhancement would provide greater flexibility in selecting alternative regularisation forms (beyond the quadratic form) and offer a more robust framework for estimating uncertainties or the significance of features reconstructed by the potential correction method.

This work represents ongoing progress in the development of free-form lens modelling tools. Over the coming years, large-scale surveys such as Euclid, CSST, and Roman are anticipated to observe hundreds of thousands of galaxy–galaxy strong lenses ([Collett 2015](#); [Cao et al. 2024](#); [Weiner et al. 2020](#); [Nagam et al. 2025](#)). These surveys may yield thousands of high-quality strong-lensing systems suitable for subhalo detection ([O’Riordan et al. 2023](#)), opening a new observational window into the properties of dwarf satellite galaxies beyond the Local Universe ([Wang et al. 2025](#)). The objective and automated

characteristics of our potential correction code render it especially suitable for analysing such extensive lens samples.

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DATA AVAILABILITY

The code and data product that supports this work are publicly available from https://github.com/caoxiaoyue/lensing_potential_correction.

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APPENDIX A: BAYESIAN FRAMEWORK FOR POTENTIAL CORRECTION

Equation 6 shows that δd and $\delta\psi$ are connected via a linear mapping. The most likely solution for $\delta\psi$ can be derived by maximizing the likelihood

$$P(\delta d | \delta\psi, \mathbf{L}_{\delta\psi}) = \frac{\exp(-E_D(\delta d | \delta\psi, \mathbf{L}_{\delta\psi}))}{Z_D}, \quad (\text{A1})$$

where

$$E_D(\delta d | \delta\psi, \mathbf{L}_{\delta\psi}) = \frac{1}{2}(\mathbf{L}_{\delta\psi}\delta\psi - \delta d)^T \mathbf{C}_D^{-1}(\mathbf{L}_{\delta\psi}\delta\psi - \delta d) = \frac{1}{2}\chi^2, \quad (\text{A2})$$

and

$$Z_D = (2\pi)^{N_d/2} (\det \mathbf{C}_D)^{1/2}. \quad (\text{A3})$$

This is equivalent to minimizing E_D with a linear least squares algorithm, which yields

$$\delta\psi_{\text{ml}} = (\mathbf{L}_{\delta\psi}^T \mathbf{C}_D^{-1} \mathbf{L}_{\delta\psi})^{-1} \mathbf{L}_{\delta\psi}^T \mathbf{C}_D^{-1} \delta d. \quad (\text{A4})$$

The most likely solution $\delta\psi_{\text{ml}}$ is usually degraded by noise and is ill-posed (Warren & Dye 2003). An additional prior $P(\delta\psi | \mathbf{g}_{\delta\psi}, \boldsymbol{\xi}_{\delta\psi})$ is needed to impose smoothness on $\delta\psi$, a process commonly called “regularisation”. Here, $\mathbf{g}_{\delta\psi}$ represents the type of regularisation, and the associated model parameters are denoted by the vector $\boldsymbol{\xi}_{\delta\psi}$. The prior can be expressed as:

$$P(\delta\psi | \mathbf{g}_{\delta\psi}, \boldsymbol{\xi}_{\delta\psi}) = \frac{\exp(-E_{\delta\psi}(\delta\psi | \mathbf{g}_{\delta\psi}, \boldsymbol{\xi}_{\delta\psi}))}{Z_{\delta\psi}(\boldsymbol{\xi}_{\delta\psi})}, \quad (\text{A5})$$

where $Z_{\delta\psi}(\boldsymbol{\xi}_{\delta\psi}) = \int d^N \delta\psi \exp(-E_{\delta\psi})$ is the normalization factor of the prior probability density function. The function $E_{\delta\psi}$, also known as the regularisation function, typically takes a quadratic form: $E_{\delta\psi}(\delta\psi) = \frac{1}{2}\delta\psi^T \mathbf{R}_{\delta\psi} \delta\psi$, where $\mathbf{R}_{\delta\psi}$ represents the Hessian of $E_{\delta\psi}$. This form allows $\delta\psi$ to be directly solved through matrix inversion (Warren & Dye 2003). For more detailed information on the specific form of the regularisation function, refer to Section 2.4.

In Bayesian inference, the posterior distribution of $\delta\psi$ is obtained by multiplying the likelihood function by the regularisation prior, as follows:

$$P(\delta\psi | \delta d, \mathbf{L}_{\delta\psi}, \boldsymbol{\xi}_{\delta\psi}, \mathbf{g}_{\delta\psi}) = \frac{P(\delta d | \delta\psi, \mathbf{L}_{\delta\psi})P(\delta\psi | \boldsymbol{\xi}_{\delta\psi}, \mathbf{g}_{\delta\psi})}{P(\delta d | \mathbf{L}_{\delta\psi}, \boldsymbol{\xi}_{\delta\psi}, \mathbf{g}_{\delta\psi})}. \quad (\text{A6})$$

Here, the normalization factor $P(\delta d | \mathbf{L}_{\delta\psi}, \boldsymbol{\xi}_{\delta\psi}, \mathbf{g}_{\delta\psi})$, known as Bayesian evidence, can be used to determine the choice of $\{\mathbf{L}_{\delta\psi}, \mathbf{g}_{\delta\psi}, \boldsymbol{\xi}_{\delta\psi}\}$ (Suyu et al. 2006, 2009). The most probable solution of $\delta\psi$ maximizes $P(\delta\psi | \delta d, \mathbf{L}_{\delta\psi}, \boldsymbol{\xi}_{\delta\psi}, \mathbf{g}_{\delta\psi})$, which is equivalent to minimizing the following penalty function:

$$M(\delta\psi) = E_D(\delta\psi) + E_{\delta\psi}(\delta\psi). \quad (\text{A7})$$

This yields Equation 9:

$$\delta\psi_{\text{MAP}} = (\mathbf{L}_{\delta\psi}^T \mathbf{C}_D^{-1} \mathbf{L}_{\delta\psi} + \mathbf{R}_{\delta\psi})^{-1} \mathbf{L}_{\delta\psi}^T \mathbf{C}_D^{-1} \delta d,$$

which indicates that, given the macro-model outputs $\{\delta d, \mathbf{L}_{\delta\psi}\}$ and an arbitrary choice of regularisation $\mathbf{R}_{\delta\psi}(\boldsymbol{\xi}_{\delta\psi}, \mathbf{g}_{\delta\psi})$, the most probable solution, $\delta\psi_{\text{MAP}}$, can be obtained via linear matrix inversion. The covariance matrix ($\Sigma_{\delta\psi}$), which describes the uncertainty of the

most probable solution ($\delta\psi_{\text{mp}}$), is derived by [Suyu et al. \(2006\)](#) and follows:

$$\Sigma_{\delta\psi} = \left(\mathbf{L}_{\delta\psi}^T \mathbf{C}_D^{-1} \mathbf{L}_{\delta\psi} + \mathbf{R}_{\delta\psi} \right)^{-1}. \quad (\text{A8})$$

For a given choice of regularisation type $\mathbf{g}_{\delta\psi}$, the values of its hyperparameters $\xi_{\delta\psi}$ can be optimally determined by maximizing the posterior $P(\xi_{\delta\psi} | \delta\mathbf{d}, \mathbf{L}_{\delta\psi}, \mathbf{g}_{\delta\psi})$, where

$$P(\xi_{\delta\psi} | \delta\mathbf{d}, \mathbf{L}_{\delta\psi}, \mathbf{g}_{\delta\psi}) = \frac{P(\delta\mathbf{d} | \mathbf{L}_{\delta\psi}, \xi_{\delta\psi}, \mathbf{g}_{\delta\psi}) P(\xi_{\delta\psi})}{P(\delta\mathbf{d} | \mathbf{L}_{\delta\psi}, \mathbf{g}_{\delta\psi})}. \quad (\text{A9})$$

Here, the evidence term $P(\delta\mathbf{d} | \mathbf{L}_{\delta\psi}, \mathbf{g}_{\delta\psi})$ marginalizes over the hyperparameters of the regularisation and can be used to rank different model choices of $\{\mathbf{L}_{\delta\psi}, \mathbf{g}_{\delta\psi}\}$. Since a non-informative prior is usually taken for $P(\xi_{\delta\psi})$, finding the most probable solution for $\xi_{\delta\psi}$ is equivalent to maximizing $P(\delta\mathbf{d} | \mathbf{L}_{\delta\psi}, \xi_{\delta\psi}, \mathbf{g}_{\delta\psi}) \equiv \mathcal{E}$, whose explicit form is given in Equation 10.

APPENDIX B: OPTIMAL HYPERPARAMETER VALUES FOR THE MATERN KERNEL

In this work, the Matérn kernel is employed to regularise the pixelised potential corrections. As defined in Equation 19, the Matérn kernel is characterised by three hyperparameters: a coefficient $\lambda^{\delta\psi}$ that controls the overall smoothing strength; a characteristic length-scale ρ (in units of arcseconds) that governs the smoothing scale; and a parameter ν that determines the order of the smoothing, ensuring the reconstructed field is $\lceil \nu \rceil - 1$ times differentiable. We assign non-informative priors to these three hyperparameters: $\lambda^{\delta\psi} \sim \mathcal{L}(10^{-6}, 10^6)$, $\rho \sim \mathcal{L}(10^{-4}, 10^3)$, and $\nu \sim \mathcal{U}(0.5, 10.0)$. The optimal values for these hyperparameters, presented in Table B1, were determined by sampling the Bayesian evidence.

APPENDIX C: ROBUSTNESS ANALYSIS OF THE POTENTIAL CORRECTION RESULTS FOR THE JACKPOT LENS

This section systematically evaluates the robustness of our potential correction method applied to the Jackpot lens. Figure C1 presents the potential correction results obtained using a larger annular mask. Comparison of the resulting convergence perturbation map with that derived using a smaller mask (Figure 10) shows that the properties of the reconstructed subhalo are consistent between the two cases. This demonstrates that our potential correction method is robust against the choice of mask size.

In addition to the localised mass clumps corresponding to the subhalo, additional mass features are observed in the bottom-right panel of Figures 10 and C1, including a dipole-like structure near the detected subhalo and a negative convergence peak at $(x, y) \approx (1.0, 1.0)$. To investigate whether these additional mass features are genuine or induced by systematic errors in our potential correction method, we adopt a parametric approach analogous to that employed by [Despali et al. \(2024\)](#) in modelling the Jackpot lens. This approach incorporates an EPL plus shear model for the mass distribution of the main lens, an NFW model for the subhalo, and a pixelized source model. Subsequently, we simulate 1000 Jackpot-like mock lenses based on this model, each with a distinct realisation of image noise. Applying our potential correction method to these mock lenses consistently reproduces the additional mass features observed in the analysis of the real Jackpot lens data. We then utilise the median and standard deviation computed from the 1000 convergence perturbation maps

(derived from the 1000 Jackpot-like mock lenses) as proxies for the signal and noise of the reconstructed mass perturbation features. The ratio between these quantities defines the signal-to-noise ratio (SNR), which quantifies the statistical significance of the reconstructed mass perturbation features. We find that the SNR of these additional mass features exceeds 3, confirming that they represent genuine artefacts induced by a systematic error in our potential correction method rather than being introduced by random noise (noting that these additional mass features should be absent in the mock lens data). If the potential correction is initiated using the true main lens mass model and source light model, the additional mass features vanish from the correction result, as illustrated in the bottom-right panel of Figure C2. We thus conclude that these additional features result from initiating the potential correction with a biased estimate of the main lens mass and source light provided by the macro model. Notably, the convergence correction shown in the bottom-right panel of Figure C2 recovers the central high-density region of the cuspy subhalo more accurately than the correction initiated from a biased estimate (see Figure 11). Hence, in addition to the limited dynamical range arising from the use of constant regularisation in the pixelized potential correction (discussed in Section 5.1), initiating the correction with a biased main lens mass and source estimation also contributes to the underestimation of the central convergence of a cuspy subhalo.

A key strength of our potential correction method is its ability not only to detect the presence of a dark matter subhalo but also to reveal its spatial mass distribution. This capability allows us to conclude, using a model-independent approach, that the subhalo in the Jackpot lens is overconcentrated, potentially challenging the standard cold dark matter (CDM) model. To assess the robustness of our potential correction method for measuring subhalo compactness, we simulated two Jackpot-like mock lenses using subhalo properties from [Despali et al. \(2024\)](#) and [Nightingale et al. \(2024\)](#), respectively, while keeping the main lens mass and source light distribution fixed. Applying our potential correction method to these mock lenses allowed us to detect a localised perturber in both cases (see Figure C3). Moreover, the detected perturber was more compact in the mock lens based on the [Despali et al. \(2024\)](#) subhalo model. This demonstrates that our potential correction method can determine the compactness of the localised perturber.

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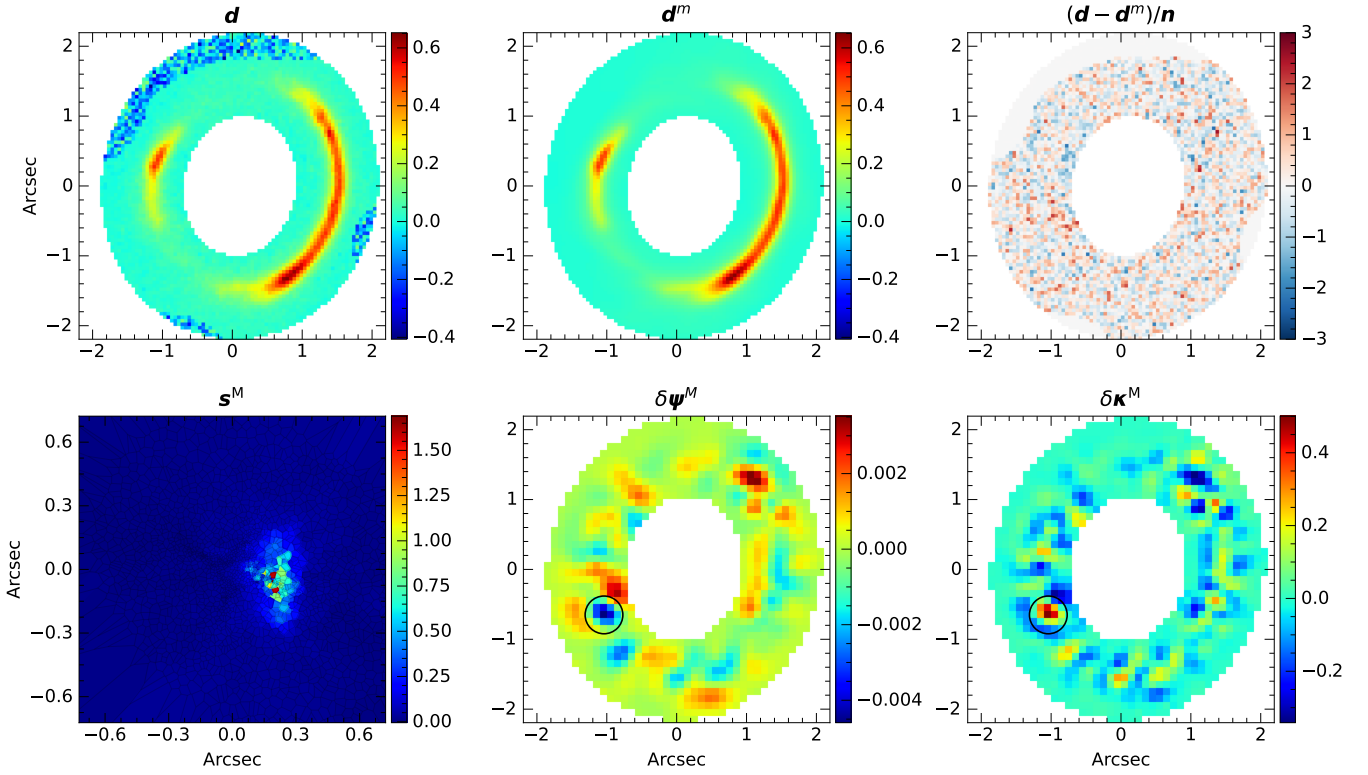


Figure C1. Similar to Figure 10, but showing the potential correction result obtained with a larger annular mask.

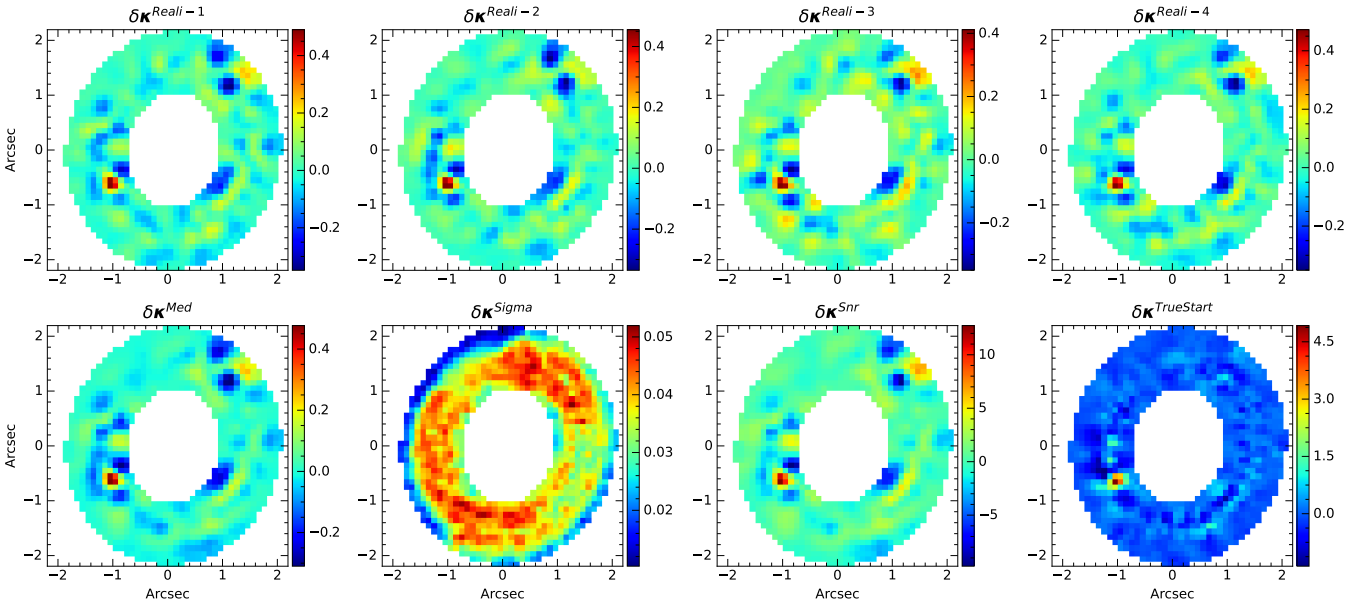


Figure C2. Assessment of perturbative convergence reconstruction using the potential correction method via Monte Carlo simulations. The top row displays convergence corrections for four mock Jackpot-like lenses differing only in noise realisation. The second row (left and middle-left panels) shows the mean and standard deviation of convergence corrections across 1000 mock Jackpot-like lenses with varying noise realisations. The signal-to-noise ratio (SNR) of the convergence correction, defined as the ratio of the mean to the standard deviation, quantifies the significance of the convergence correction. The bottom-right panel illustrates the convergence correction for a mock Jackpot-like lens initialised with the true main lens mass and source light distributions.

		log[Coefficient]	log[Scale]	Nu
Spherical NFW Subhalo	$\delta\psi$ -only	$3.1919^{+0.3860}_{-0.5423}$	$0.6997^{+0.1881}_{-0.1426}$	$2.2389^{+0.3510}_{-0.2562}$
	$(\delta\psi, s)^{\text{Truth}}$	$3.5091^{+0.4556}_{-0.6482}$	$0.5136^{+0.2297}_{-0.1791}$	$2.8750^{+1.9273}_{-0.6309}$
	$(\delta\psi, s)^{\text{Macro}}$	$4.9545^{+0.2624}_{-0.3180}$	$0.0182^{+0.0877}_{-0.0737}$	$6.5831^{+2.3161}_{-2.3562}$
Shear	$\delta\psi$ -only	$2.0746^{+0.1259}_{-0.0556}$	$0.7473^{+0.0712}_{-0.0578}$	$5.5018^{+3.0553}_{-2.1787}$
	$(\delta\psi, s)^{\text{Truth}}$	$2.0900^{+0.1322}_{-0.0662}$	$0.7020^{+0.0628}_{-0.0590}$	$7.2722^{+1.8435}_{-2.0310}$
	$(\delta\psi, s)^{\text{Macro}}$	$3.5641^{+0.2972}_{-0.3101}$	$0.2702^{+0.0547}_{-0.0551}$	$8.3728^{+2.1427}_{-1.8146}$
Multipole	$\delta\psi$ -only	$4.3240^{+0.1558}_{-0.1930}$	$0.0260^{+0.0733}_{-0.0523}$	$5.6456^{+2.2337}_{-1.4727}$
	$(\delta\psi, s)^{\text{Truth}}$	$4.1447^{+0.2028}_{-0.2470}$	$0.0518^{+0.0845}_{-0.0572}$	$6.1994^{+2.3352}_{-1.9377}$
	$(\delta\psi, s)^{\text{Macro}}$	$4.2074^{+0.1840}_{-0.2042}$	$-0.0091^{+0.0805}_{-0.0542}$	$6.1800^{+2.4526}_{-2.0224}$
Gaussian Random Field	$\delta\psi$ -only	$2.6879^{+0.4094}_{-0.4496}$	$0.7515^{+0.1646}_{-0.1663}$	$1.7675^{+0.1484}_{-0.1261}$
	$(\delta\psi, s)^{\text{Truth}}$	$2.8467^{+0.4500}_{-0.5086}$	$0.6596^{+0.1992}_{-0.1939}$	$1.8761^{+0.2691}_{-0.1987}$
	$(\delta\psi, s)^{\text{Macro}}$	$4.4075^{+0.2332}_{-0.3383}$	$0.0351^{+0.1423}_{-0.0869}$	$3.8699^{+2.8851}_{-1.2515}$

Table B1. Optimal hyperparameter values for regularising lensing potential perturbations using the Matérn kernel, determined objectively by sampling the Bayesian evidence. Results are shown for inversions of the lensing potential only ($\delta\psi$ -only), joint inversions of the source and lensing potential using the true main lens and source models $((\delta\psi, s)^{\text{Truth}})$, and joint inversions initialised from the point estimate of the macro model without potential perturbations $((\delta\psi, s)^{\text{Macro}})$. Superscripts and subscripts represent the 68% credible interval.

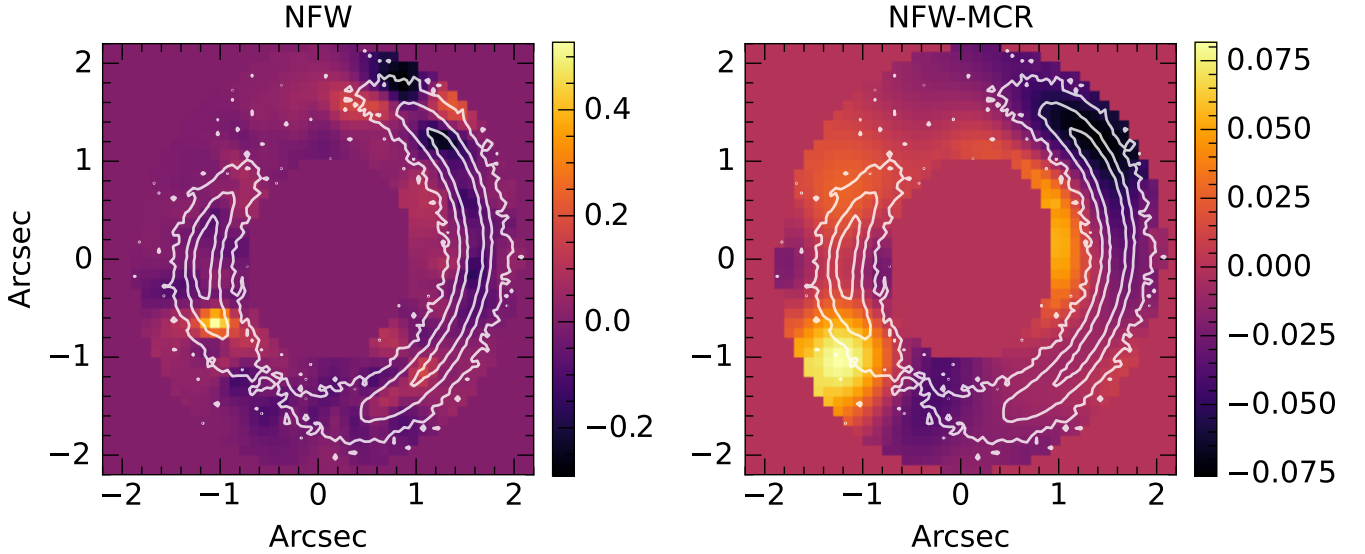


Figure C3. Convergence correction results for two Jackpot-like mock lenses, demonstrating the ability of our potential correction method to reveal the mass distribution of the input subhalos. The left lens includes an input NFW subhalo with a mass of $\sim 10^{10} M_{\odot}$ and a high concentration (~ 200), centred at $(-1.040, -0.651)$ (Despali et al. 2024). The right lens contains an input NFW subhalo with a mass of $\sim 10^{11.5} M_{\odot}$, following the standard mass-concentration relation, centred at $(-1.00, -1.10)$ (Nightingale et al. 2024).